

Recursive Axis Conditioning for Diverse Synthetic Data Generation

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Abstract

A synthetic corpus is judged by a measure, and the measure is chosen before the corpus is. **Recursive Axis Conditioning** (RAC) is a generation loop that improves a *chosen* measure: it asks the generator to name the language-valued axes along which its own outputs can differ, ranks them, takes their most-different levels, and splits an exhausted axis into finer sub-axes. The measure enters at exactly two points — how an axis is scored and how one of K candidates is selected — and everything else is shared.

It wins where it is pointed. On coverage of a held-out human-written reference that no corpus was aimed at, RAC places **first of twelve corpora, 0.4441 against Alpaca's 0.3722** at matched evaluated n , from a corpus built with 2,400 generator calls against Alpaca's 52,002 released items, with the highest precision in the field (0.973); PersonaHub (50k) and WizardLM Evol-Instruct (143k) finish below RAC's 304-item selective arm. On automatic item generation, a naively prompted bank of 2,500 items yields **382 that can legally coexist on one exam form — 15.3% of nominal capacity**, against 94.9% for human-written MMLU, while RAC's 1,999-item bank contains **no enemy pair at all** at a blind-adjudicated radius.

The evidence we trust most is measured in channels the loop never optimizes, so it cannot be a restatement of the selection rule. At matched generator budget, RAC separates from the strongest available comparison in all three artifact domains: **2.25× the prosodic spread** of naive prompting on 200 of 200 random draws (2.02× with every length feature removed); **97.6% distinct item architectures against 43.9%** after deleting every exact duplicate from both banks; and **1.53× the optical spread** on pixel statistics, [1.38, 1.68], with every one of eleven RAC image corpora above every one of five published baselines ($U = 55$ of 55, $p = 0.0002$).

The first result to generalize past the method is that **corpus objectives divide into two classes and the machinery load-bearing for one is harmful to the other**. Greedy k -center, the classical packing algorithm, wins min-gap in both of our domains and finishes last on coverage, at a sixth of random selection; orthogonalized conditioning, which is what makes RAC work under max-min, finishes last of the nine seeded arms on coverage and below plain conditioning at both matched n and matched budget; the two highest-Vendi selectors are the two worst covering ones. A diversity number carries no information until its objective is named.

Second, a **completeness rule** for conditioning, established over seven controlled comparisons with blind judges. An axis that enumerates a proper subset of the space a measure is defined over displaces concentration into the complement rather than removing it; only enumerating the whole space reduces it. Naming five of ten task categories drove the unnamed five from 86% of a corpus to 19% and changed total

spread not at all (-0.014 , 95% CI $[-0.080, +0.058]$); naming all ten raised spread in a lumpy corpus ($+0.219$, $[+0.138, +0.297]$) and in an already-flat one ($+0.146$, $[+0.069, +0.225]$). Inside a single level the same rule holds: an axis enumerating every form a red herring can take moved an item bank's dominant device from 76% of items to 24% (-0.507 , $[-0.622, -0.389]$) while both arms kept the contract at 100%, where axes naming only part of the realization space moved poems by 0.04 or made them worse.

Underneath all of it is one limit. We assume an embedding oracle and no inverse — we can compute where the next item ought to land and cannot decode that point into text — so steering is language-valued, and no selection rule can exceed what the conditional support offers. Fifteen numerical checks confirm that novelty at a fixed prompt decays as $n^{-1/m}$ in the conditional dimension rather than $n^{-1/d}$ in the ambient one, that a single prompt ϵ -covers a vanishing ϵ^{d-m} fraction, and that only prompt motion transverse to the occupied span raises the ceiling. Evidence: over 46,000 real generations, $\sim 1,800$ rendered images and ~ 200 rendered instrumentals, for roughly \$40 of API spend.

1. Introduction

Ask a language model for a poem ten times and you get ten poems. Ask it ten thousand times and you get a few hundred poems and a great deal of paraphrase. Any fixed conditional distribution behaves this way under repeated sampling, whatever model supplies it: the distribution has a shape, and sampling traces that shape ever more densely rather than expanding it.

The practical version is now everywhere. Synthetic training data is worth what it adds to the training set; an evaluation suite that clusters in a few families gives false assurance; test-item banks, red-team prompt sets, persona corpora and augmentation pipelines all consume generated text in bulk. All of them degrade in a way that per-item quality checks do not detect — every item is fine, and the collection is redundant — and the corpora shipped today have it badly. Asked a reasonable psychometric question ten thousand times, a strong model returns a bank that is **73.6% byte-identical duplicates**, with one question repeated 2,726 times; of a 2,500-item sample only 382 items can legally coexist on one exam form.

This paper presents **Recursive Axis Conditioning (RAC)**, a generation loop that fixes this well enough to beat the best released corpus with one a twentieth of its size, and three results that came out of building it.

The loop works, and the margin does not come from scale. Against Alpaca, PersonaHub and WizardLM Evol-Instruct on coverage of a held-out human reference, RAC places first of twelve corpora from 2,400 generator calls, against Alpaca's 52,002 released items (§7.3). On psychometric item generation it produces a bank with no enemy pair at all under a blind-adjudicated radius, where the naive bank yields 15.3% of its nominal capacity and human-written MMLU yields 94.9% (§7.4). And the separation survives being measured where the method cannot be gaming it: in each of the three artifact domains there is a channel nothing in the loop optimizes — prosody for poems, item architecture for exam banks, pixel statistics for renders — and RAC separates from the baselines on all three at matched generator budget (Figure 1, §7.5).

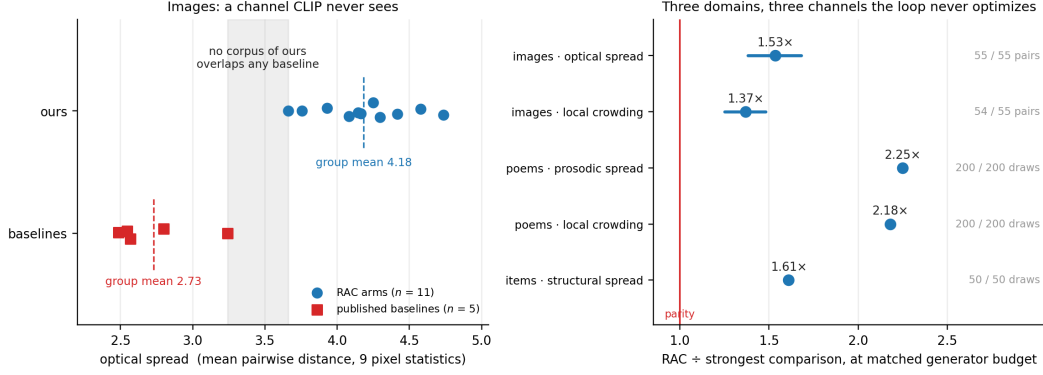


Figure 1. Separation measured outside the objective. Left: every image corpus at matched $n = 60$ on nine pixel statistics — a channel with no connection to the CLIP or text embeddings the loop selects on. Every one of our eleven corpora exceeds every one of the five published baselines. Right: the same comparison in all three artifact domains, on the channel each domain’s objective cannot see, at matched generator budget; intervals are bootstrap 95% CIs on the ratio of group means where the group sizes support them, and the annotation gives the number of pairs or draws favouring RAC.

The objective has to be named before the number means anything. Corpus objectives fall into two classes — *covering* the reachable space, and *packing* it so no two items collide — usually spoken of interchangeably. They are not interchangeable, and each class’s characteristic tool damages the other’s score. Greedy k -center, the classical packing algorithm, wins min-gap in both of our domains and finishes last on coverage, at a sixth of random selection. Orthogonalized conditioning, the component that makes RAC work under packing, *reduces* coverage below plain conditioning when carried across, and finishes last of the nine seeded arms. The two highest-Vendi selectors we measure are the two worst covering ones (§3).

Conditioning relocates a corpus’s repetition unless it names the whole space the measure is defined over. This is the prescription we would keep if we could keep one. An axis enumerating five of ten task categories moves an enormous amount of probability mass onto those five and leaves total spread unchanged, because the five it does not name are driven out; an axis enumerating all ten raises spread in both a lumpy corpus and an already-flat one. The same rule holds one level down, inside a single axis level, where a construct realized through one lexical device is invisible to every realization audit in the literature and to every diversity measure in this paper (§8.3).

Underneath all three is a single limit, made precise in §4: no selection rule can outrun the conditional support of the generator it draws from, so the only move that changes the asymptote is one that widens that support.

1.1 The oracle asymmetry

We assume throughout:

Assumption (embedding oracle, no inverse). We have access to $E: \text{Text} \rightarrow \mathbb{R}^D$, computable on demand. We have **no** $E^{-1}: \mathbb{R}^D \rightarrow \text{Text}$.

This is the structural fact that shapes the entire design. Given a corpus X_n we can compute, exactly and cheaply, the point in $\mathbb{R}^D$ that would most improve any of our diversity measures — the direction of least occupied spectral energy, the centre of the largest empty ball, the point maximizing marginal coverage. Knowing that point is worth nothing on its own, because no procedure turns a target embedding back into a poem.

Consequently the system cannot *solve* for its next item. It can only **propose** (sample from $p(\cdot | x)$ for some prompt x we can write), **measure** (embed the proposals and score them), and **select** (keep one). All of the control is in the choice of x , and that choice is expressible only in language. This is why the latent variables here are language-valued — named axes with named levels — rather than continuous codes: they are the only handles that reach the generator. It is also why the axis scoring of

§5 exists. **The axis scoring is a surrogate for the missing inverse:** unable to decode the direction we want to travel, we score the language-valued conditions we *can* write by how nearly their induced output distributions point that way.

1.2 Contributions

- **A generator loop that improves a chosen measure** (§5). Language-valued axes elicited from the generator, ranked by a scoring rule, with exhausted ones split recursively into conditional sub-axes. The measure enters at two points only, so pointing the loop at a different measure means changing those two and nothing else.
- **A competitive result across five domains** (§7). Against five published methods at matched n , RAC wins every metric in both text domains, beating the strongest baseline by 36% in mean-centered Vendi at matched $n = 1,000$; its exam bank contains no enemy pair at the judged radius, against 94.9% usable for human-written MMLU and 15.3% for naive prompting; and against released instruction corpora it places first of twelve at 0.4441 against Alpaca's 0.3722.
- **Separation on channels the method never optimizes** (§7.5), in all three artifact domains, at matched generator budget, with the most obvious confound removed in each case. These are the comparisons that cannot be a restatement of the selection rule, and they are the ones we would defend first.
- **Evidence that the measure has to be named** (§3). Greedy k -center wins min-gap in both domains and finishes last on coverage, and each measure's characteristic tool damages the other's score.
- **A manipulation-based scoring rule for conditioning variables** (§5.3). The obvious observational estimator is *inverted*: it penalizes an axis in proportion to how well the axis works. On data where the answer is fixed by construction, the observational form ranks the inert axis first in 17 of 24 runs; the repair recovers the true order in 24 of 24.
- **The completeness rule for conditioning** (§8.3), with the two audits that make it visible: realization has to be checked between levels *and* within one, and an axis can be present, obeyed, and still realized the same way every time.
- **A theory of what limits both objectives** (§4), with fifteen numerical checks.

What is deferred to a companion paper. Every measure here is computed in an embedding, and in two of our domains that embedding is a proxy for an artifact that lives elsewhere — a rendered image, a rendered instrumental. How large the gap between proxy and artifact is, which embedder to steer through, how to close the loop in the artifact's own space, and what an embedding cannot see at all are treated in *Proxy Embeddings: Steering and Measuring Diversity Through a Space That Is Not the Artifact's*, which extends the method across that gap. Where this paper depends on one of those results it says so and cites the section.

2. Related work

Synthetic instruction corpora. Self-Instruct (Wang et al., 2023) bootstraps instructions from a seed pool with few-shot exemplars and a ROUGE-L similarity filter; the Stanford Alpaca corpus (Taori et al., 2023; 52,002 items) is its canonical output. Evol-Instruct / WizardLM (Xu et al., 2024) *mutates* existing instructions with depth and breadth operators, released as WizardLM_evol_instruct_V2 (143,000 items). Persona-Hub (Ge et al., 2024) conditions on a catalogue of ~1B personas, releasing 50,000 persona-synthesized instructions; it is the closest published relative of latent-axis conditioning, differing in that its catalogue is mined from web data and flat, where our axes are elicited from the generator and refined into a tree. AttrPrompt (Yu et al., 2023) similarly conditions on attribute dimensions. We compare against the *released artifacts* of the first three (§7.3) rather than against reimplementations, since a reimplementation can be weak in ways that flatter us.

Selection and subset choice. Farthest-point traversal (Gonzalez, 1985) 2-approximates k -center and, by the same argument, max-min dispersion, the packing objective (Ravi et al., 1994); SemDeDup (Abbas et al., 2023) removes semantic near-duplicates at a fixed radius; determinantal point processes (Kulesza & Taskar, 2012; Chen et al., 2018) model repulsion via volume. All are *post hoc* selectors over a fixed pool, which is their limitation here: they cannot change what is in the pool, and §4 shows the pool is the binding constraint. We use all three as baselines.

Submodular maximization and the packing/covering distinction. The $(1 - 1/e)$ greedy guarantee for monotone submodular objectives is Nemhauser, Wolsey & Fisher (1978); hardness of beating it for max-coverage is Feige (1998). Sieve-streaming (Badanidiyuru et al., 2014) gives the single-pass regime our sequential setting approximates, and facility location (Lin & Bilmes, 2011) is the standard submodular model of representativeness. k -median/facility-location minimizes *average* service distance and behaves like our coverage objective — measure-weighted, bulk-first, outlier-indifferent — where k -center does not. Our contribution is not new theory for these objects but the identification of which one a corpus-construction requirement is, and measurement of what optimizing the wrong one costs (§3.4).

Quality-diversity. MAP-Elites (Mouret & Clune, 2015) maintains one elite per cell of a hand-designed behavior space. Our axis lattice is a close relative with three differences that matter: the descriptors are elicited from the generator rather than designed by us, the lattice refines itself when a cell saturates, and allocation is marginal-gain-driven rather than one-per-cell. §4.6 quantifies what the fixed-grid choice costs at small ϵ .

Diversity metrics. The Vendi Score (Friedman & Dieng, 2023) is the exponential of the von Neumann entropy of a normalized similarity matrix, interpretable as an effective number of distinct items; we use it throughout, and §7.1 and §7.5 record two regimes in which it is the wrong estimator. Alignment narrows what a model writes: RLHF lowers output diversity relative to supervised fine-tuning across a range of measures (Kirk et al., 2024), and we take that narrowing as the empirical starting point. Red-teaming work clusters the attacks it collects by type (Ganguli et al., 2022) and finds a trade-off between how often generated test cases succeed and how diverse they are (Perez et al., 2022), which is the case for a coverage objective in safety suites.

3. Two objectives, and why the number needs one named

Covering and packing are routinely treated as two phrasings of one goal. On real corpora they rank methods almost oppositely, and each one’s characteristic tool costs the other measurably. This section states both, measures the reversal, and draws the consequence for the loop.

3.1 Packing (max-min)

Under an unbounded horizon the objective is a two-term score evaluated against everything generated so far. Let E be an embedding oracle and $X_n = \{x_1, \dots, x_n\}$ the corpus. For a candidate c ,

$$J_\lambda(c \mid X_n) = \underbrace{\lambda \cdot \frac{1}{n} \sum_{i=1}^n \|E(c) - E(x_i)\|}_{\text{anchor: keep it typical}} - \underbrace{(1 - \lambda) \cdot \min_{i \leq n} \|E(c) - E(x_i)\|}_{\text{repulsion: keep it new}}$$

and we accept the candidate minimizing J_λ . The anchor keeps generation on the manifold, the repulsion pushes away from what exists. Both terms are cheap online: the anchor decomposes into distance-to-centroid plus a candidate-independent constant, so ranking by it is $O(D)$ per candidate and $O(1)$ in n , and the repulsion needs one nearest-neighbour query, bounded by subsampling. Packing is the right question when a single collision is a defect on its own — two exam items testing the same rule are a security failure whatever the rest of the bank looks like.

3.2 Coverage

Fix an embedding map into $\mathbb{R}^D$ (we use unit-normalized 768-d embeddings) and a radius $\epsilon > 0$. The generator, prompted in whatever ways the pipeline can reach, induces a *reachable distribution* μ . Given a budget n , choose $S = \{x_1, \dots, x_n\}$ to maximize the covered measure

$$F_{\mu, \epsilon}(S) = \mu\left(\bigcup_{x \in S} B(x, \epsilon)\right),$$

the probability that a fresh draw lands within ϵ of some chosen item. Equivalently, S should be an ϵ -net for as much of μ as the budget allows. Coverage is the right question when the corpus stands for a population, as an evaluation suite or a training set does. Three choices are deliberate:

- **Coverage of the reachable space, not of $\mathbb{R}^D$.** Balls in empty space cover volume but no behaviour. The denominator is stated next to every coverage number; where corpora built by

different methods are compared it is a held-out human-written reference identical for all of them (§7.3).

- **ϵ is a resolution parameter, not a nuisance.** Small ϵ asks for exemplars of fine distinctions, large ϵ for one exemplar per coarse region, and the ranking of methods changes with ϵ . Every result names its radius or range of radii.
- **Quality is a constraint, not part of the objective.** An item that covers new territory but fails a quality bar is not an asset, so the judge gate rejects it regardless of gain.

3.3 Where they coincide, and where they do not

For a fixed ϵ , a *maximal* ϵ -packing is automatically an ϵ -covering: any uncovered point could have been added to the packing. We confirm this numerically on 3,000 held-out human instructions — at every ϵ tested, the greedy maximal packing covers 100.0% of the set, and the classical sandwich $N_{cov}(\epsilon) \leq N_{pack}(\epsilon) \leq N_{cov}(\epsilon/2)$ holds. Run to maximality, max-min is a covering algorithm.

That duality concerns the worst-case radius. The coverage that matters for synthetic data is **measure-weighted**, and the two come apart exactly when the measure is non-uniform, which it always is. k -center is driven by outliers: every isolated point sets the max and so commands a centre, and at a budget far below what the space needs, every such item is a ball spent covering ≈ 0 measure. Measure-weighted coverage is driven by mass: a second exemplar in a heavy mode can be worth more than a first exemplar of a negligible one, and junk is worth nothing because μ puts almost nothing near it. The true dual of measure-weighted coverage is fractional set cover, whose primal is submodular; max-min is not.

What the duality still buys is calibration. Even where packing is the wrong way to *place* points it is the right way to choose the *scale*: $\epsilon^*(n)$, the radius at which a maximal packing of the reference has exactly n centres, is the largest resolution at which n items could in principle cover, and it is the radius we calibrate to.

3.4 The reversal, measured

A leakage-free selection benchmark isolates the contrast with no generator in the loop: one shared candidate pool, budget matched at 700 and 1,896 generations, selector shown only an estimation half of the reference and scored on a disjoint held-out half.

selector	DALL·E coverage	psychometric coverage	DALL·E min-gap
coverage-greedy (ours)	0.4829	0.3901	0.299
coverage-stream (ours)	0.4200	0.3785	0.283
ROUGE-L filter (Self-Instruct)	0.4629	0.3575	0.208
SemDeDup	0.4600	0.3575	0.336
random	0.4629	0.3533	0.198
MAP-DPP greedy	0.3829	0.0988	0.402
k -center (Gonzalez)	0.4114	0.0557	0.491

k -center wins min-gap outright in both domains and finishes last on coverage, at 0.0557 against random's 0.3533 in the psychometric domain — a sixth of what picking at random achieves. The two highest-Vendi selectors, k -center and MAP-DPP, are the two worst covering ones. A practitioner who reads "diversity" off a Vendi score and deploys the selector that maximizes it gets a corpus covering less of the space than random selection.

The same dissociation appears with the full generation loop in place, and in the other direction. The configuration that applies the packing side's *orthogonalized conditioning* to the coverage objective finishes **last of the nine seeded arms** in §7.3, below plain conditioning at matched evaluated n (0.2102 against 0.2127) and by more at matched budget (0.2941 against 0.3147, though those two arms kept 1,152 and 1,310 items and the estimator is monotone in items, so part of that larger gap is corpus size). Orthogonalization steers generation away from the occupied span, which is away from the reference-dense core coverage is paid to fill. And the winning coverage corpus has the **lowest** mean-centered Vendi of any configuration in its cohort (62.1, density 1.69) — it spends items where

the reference measure is, near-duplicates included, exactly as facility location prescribes. A single "diversity score" would rank the coverage winner last.

3.5 Why the scoring rule has to fork

Two of the loop's factors invert between the classes, and the rest do not.

	packing / max-min	covering / coverage
the ask	no two items alike, in n turns	reach as much as possible, in n turns
governed by	the closest pair	the bulk
submodular	no	yes, so greedy carries $(1 - 1/e)$
classical kin	k -center	facility location
spread(a)	min distance between level centroids	measure-weighted mean distance
headroom(a)	levels not yet used	measure not yet covered
value selection	max-min subset of levels	greedy marginal gain over levels
failure mode	chases outliers into off-manifold junk	leaves the frontier empty

The behavioural difference shows on a two-line test. Given an axis with three tightly clustered common levels and one rare outlying level, max-min value selection picks the outlier **first** and coverage value selection picks a cluster centre first and the outlier second. That is k -center versus facility location reproduced inside the axis scoring, and it is the mechanism behind the benchmark above.

3.6 What both classes share, and what follows

Both are bounded by the same thing, and neither optimizer can fix it. Coverage of a reachable manifold and packing within one are both limited by that manifold's dimension, which is set by how far the prompt can move the generator (§4). Both need the same typicality constraint to be well-posed, since a novelty-seeking score can otherwise be satisfied by output that has left the manifold. And both are read through an embedder whose geometry can invert the result: we report a cross-corpus comparison in which our worst corpus by every other measure (19.8% exact duplicates) scores the *highest* coverage, because at an ϵ in the 2nd percentile of reference distances the metric rewards centrality rather than spread.

Five practices follow, and we use them throughout.

1. **Say which objective you mean.** They are different problems with different optimal policies, and "diverse" does not distinguish them.
2. **Count exact duplicates before computing anything.** A 74% duplicate rate makes every distance statistic a statistic about duplication.
3. **Publish the pairwise-similarity distribution** your kernel operates on. Mean pairwise cosine was 0.883 in one of our corpora and 0.444 in another under the same embedder; an ϵ or a Vendi score means nothing without it.
4. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven rendered arms, and 8.2% within-arm (§7.7; companion §7.1).
5. **Audit in a channel the objective never optimizes** (§7.5, §8.3). It is the only measurement in this paper that cannot be a restatement of the selection rule, and it is where both the strongest positive result and the largest undetected defect were found.

4. What limits both: the conditional slice

Nothing in this section depends on which measure was chosen. The results bound what a fixed prompt can reach at all, and a measure computed on the resulting corpus inherits that bound whatever it is measuring. The first group concerns the conditional support and applies to any selection rule; the second concerns the covering functional in particular, and is what makes greedy selection defensible when coverage is the measure. Every claim is checked numerically; §4.6 reports the checks, all fifteen of which are reproducible from the released code.

4.1 The reachable manifold and the conditional slice

Let $M \subset \mathbb{R}^D$ be the reachable semantic manifold, $\dim M = d$: the embeddings of texts the generator could produce under *some* prompt. For a fixed prompt x , let $S_x \subseteq M$ be the support of $E_{\#}p(\cdot | x)$, with $\dim S_x = m$. The empirical claim behind everything below is that $m \ll d$. A prompt fixes topic, stance, form, register and rhetorical strategy; what remains free is a low-dimensional residue. Our live pilot supports this directly: sixty poems from an identical naive prompt have a median nearest-neighbour cosine similarity of 0.911 and a Vendi Score of 2.55 — sixty samples behaving like two and a half distinct items. We write g_n for the min-gap of a fresh draw against a corpus of size n .

4.2 Saturation, and why oversampling is not a strategy

Theorem 1. Let X_n be n i.i.d. draws from a distribution with density bounded above and below on an m -dimensional manifold S . For a fresh draw c from the same distribution, $\mathbb{E}[g_n] = \Theta(n^{-1/m})$.

Proof sketch. $\mathbb{P}(g_n > r) = (1 - \mu(B(c, r)))^n$ and $\mu(B(c, r)) = \Theta(r^m)$ by density bounds on an m -manifold, so $\mathbb{P}(g_n > r) \approx \exp(-n\Theta(r^m))$, which transitions at $r = \Theta(n^{-1/m})$; integrating the tail gives the expectation. ■

The content is the exponent. With $d = 32$ a naive reading predicts $n^{-1/32}$ — essentially flat, novelty never exhausted. The true rate at $m = 3$ is $n^{-1/3}$, roughly a tenfold loss of headroom per thousandfold of corpus. Measured slopes: -0.472 ($m = 2$), -0.309 ($m = 3$), -0.195 ($m = 5$), against -0.5 , -0.333 , -0.2 .

Corollary 1.1 (oversampling is not a strategy). Drawing K candidates and keeping the most novel multiplies the expected gap by $\Theta(K^{1/m})$ at best. To recover the headroom lost by a $1000\times$ larger corpus, K must grow by 1000.

Verified at $m = 3$: best-of-8 yields 2.19 against $K^{1/3} = 2.00$. Selection is a constant-factor fix to an asymptotic problem. §8.4 measures what this ceiling looks like in a live loop, and finds that a system can fail to reach even it by pointing its selector somewhere else.

Theorem 2 (the slice deficit). If $m < d$, then $\text{vol}_d(S_x) = 0$, and the fraction of M within ε of S_x is $\Theta(\varepsilon^{d-m})$.

Proof sketch. The ε -neighbourhood of an m -dimensional set in d dimensions is a tube of volume $\Theta(\varepsilon^{d-m}) \cdot \text{vol}_m(S_x)$ for ε below the reach of S_x ; divide by $\text{vol}_d(M)$. ■

Measured at $d = 8$, $m = 2$: fitted exponent 5.06 against a predicted 6, with covered fractions falling $0.051 \rightarrow 0.0001$ as ε goes $0.5 \rightarrow 0.12$. The practical reading is severe. **A single prompt, sampled infinitely often, covers essentially none of what the model could write** — not “less than we would like”, but a fraction going to zero polynomially as the resolution of interest sharpens.

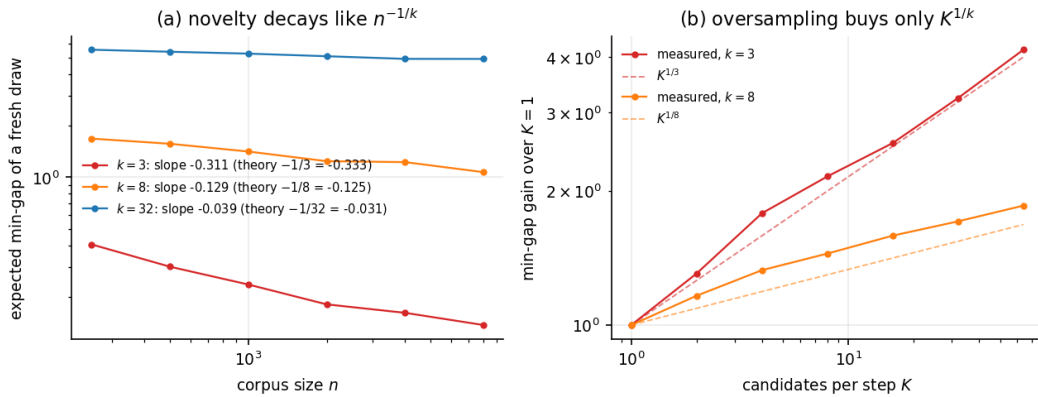


Figure 2. Measured saturation. (a) The expected min-gap of a fresh draw decays as $n^{-(1/m)}$ in the conditional dimension, with fitted slopes matching theory at $m = 3, 8, 32$. (b) Best-of- K oversampling buys only $K^{1/m}$: at the conditional dimensions we measure, quadrupling the candidate pool buys

tens of percent, not multiples.

4.3 Transversality: which prompt motion helps

Let a prompt schedule induce slices $S_1, \dots, S_P$.

Theorem 3. $\dim(\bigcup_j S_j) = \min(d, m + \text{rank}\{c_j - c_1\})$, where c_j is the centre of S_j . In particular, displacements lying inside $\text{span}(S_1)$ contribute nothing to the union's dimension.

Proof sketch. The union lies in the affine hull of the slice frame plus the span of the centre displacements; dimensions add up to that cap, and a displacement already inside the slice's own span adds no new direction. ■

Measured via the participation ratio of the union's covariance spectrum: parallel displacements give effective dimension 1.88 — the slices lie on top of each other at $m = 2$ — and transverse displacements 11.18. This is the theorem that makes the axis scoring of §5 possible: it turns "how much does conditioning on this variable buy" into **how much of its variation lies outside what the corpus already spans**, which is computable, where "how different does it sound" is not.

Theorem 4 (breadth versus depth). Given budget B split as P prompts $\times n$ samples, with per-prompt switching cost c so that $B = P(c + n)$: with $c = 0$ the coverage-optimal depth is $n^* = 1$, and for $c > 0$ the optimum is interior and increases with c .

Measured: with free switching, depth 1 achieves coverage 0.995 while spending the whole budget on one prompt achieves 0.091, a 10.9 $\times$ difference; with $c = 3$, $n^* = 10$; with $c = 30$, $n^* = 30$. **The ratio of prompt-switching cost to sampling cost sets the batch size, and nothing else does.** Our pipelines sit near the cheap-switching regime — in the live pilot we measure $c \approx 0.10$ generation-equivalents — which is why every policy here draws its K candidates from K fresh specs rather than sampling any spec deeply.

For the packing objective this leaves the two terms of J_λ cheap and the candidate set expensive. Writing μ_n for the corpus centroid, $\frac{1}{n} \sum_i \|c - x_i\|^2 = \|c - \mu_n\|^2 + \frac{1}{n} \sum_i \|x_i - \mu_n\|^2$ (verified to zero relative error), and the second term does not depend on c , so ranking by mean squared distance to the corpus is ranking by distance to the centroid — $O(D)$ per candidate, $O(1)$ in n . **The bottleneck is that both terms are evaluated on candidates drawn from an m -dimensional slice.**

4.4 Coverage: submodularity, greedy, and the streaming case

For any measure μ and radius ε , $F(S) = \mu(\bigcup_{x \in S} B(x, \varepsilon))$ is monotone (adding a ball cannot uncover anything) and submodular: the marginal gain of x given S is $\mu(B(x, \varepsilon) \setminus \bigcup_{y \in S} B(y, \varepsilon))$, and since $S \subseteq T$ implies $\bigcup_S B(y, \varepsilon) \subseteq \bigcup_T B(y, \varepsilon)$, the set being subtracted only grows. Nothing about balls is essential — any map from items to measurable footprints gives a submodular F . The same argument applies verbatim to the Monte-Carlo estimator we optimize: with a reference pool P , item x covers the fixed subset $N_\varepsilon(x) = \{p : \|p - x\| \leq \varepsilon\}$ and $\hat{F}(S) = |\bigcup_{x \in S} N_\varepsilon(x)|/P$ is a finite coverage function, monotone submodular exactly, at every sample of the pool. Greedy therefore satisfies $\hat{F}(S_{\text{greedy}}) \geq (1 - 1/e) \cdot \max_{|S| \leq n} \hat{F}(S)$, tight for the class (Feige, 1998).

A generation pipeline cannot pick from a precomputed ground set: candidates arrive K at a time and must be accepted or discarded immediately. Our per-step rule — take the best of K fresh candidates by marginal gain — is the natural best-of- K relaxation of sieve-streaming: weaker in the worst case, stronger on exchangeable streams like ours, where each batch is i.i.d. from the current proposal distribution and thresholding would only discard budget. What matters for scaling is that the marginal-gain oracle is $O(K \cdot P)$ per step against a fixed-size pool with an incrementally maintained covered mask — $O(1)$ in n .

Estimation error is Hoeffding: a 95% band of $t_{95} = \sqrt{\ln(2/0.05)/(2P)}$, about ± 0.030 at $P = 2,000$ and ± 0.015 at $P = 8,000$. That bound is for *fixed* S ; a set selected by optimizing $\hat{F}$ on the same pool is biased upward on that pool, which is why every coverage number reported here is computed on held-out material the selector never saw.

4.5 The reachability gap: conditioning, not selection, is where the coverage is

The single most consequential simulation is the one that separates the two levers. Identical greedy selection ($\epsilon = 1.0$, $k = 300$, $|\text{ground}| = 2,000$) is run against three proposal distributions and scored on one held-out full-mixture pool. An **oracle** ground set drawn from the full latent mixture — the best a selector could do if conditioning reached everything — covers 0.387. A **proposal-limited** ground set drawn from the base specs, which reach 25 of 60 modes, covers 0.212. That 17.4-point gap is one no selection algorithm can close, because the deficit is in what the generator can be made to emit. A **refined-proposal** ground set, one spec per mode, covers 0.374 and recovers 92.5% of it.

Two corollaries of the same experiment sharpen what "widening the support" means. **Lattice size is not reachable dimension**: two worlds with identical lattice size (1,024 distinct specs) but different rank of the level-direction span cover their own reachable pools at 49.0% (rank 10) against 68.1% (rank 2) — the number of distinct prompts you can write says nothing about how much there is to cover, and a huge lattice over low-rank level directions is a thin space wearing a combinatorial costume. And **refinement pays only if the children move**: tight children at $0.25\times$ displacement produce draws that are 36.7% already covered by the pre-refinement corpus and add +0.02 effective dimensions — almost pure density — where full-length isotropic displacement produces 0% pre-covered draws and +0.13, and transversality-guided displacement 0% and +0.14. Splitting a saturated cell into tighter sub-levels is worth its cost only when the split displaces.

4.6 Numerical verification

Each claim is checked against a numerical experiment designed to break it.

check	result
submodularity, exactly (5,000 nested chains, 200 candidates, 5,000-point pool)	0 violations of diminishing returns, 0 of monotonicity
greedy against <i>exact</i> optima (10 instances, 142,506 subsets enumerated each)	greedy/OPT = 1.0 on all ten; the 0.632 bound never approached
Monte-Carlo band (200 pools per size against a 200k ground truth)	95th-ptile error 0.042 / 0.020 / 0.011 at $P = 500 / 2,000 / 8,000$ against Hoeffding 0.061 / 0.030 / 0.015
saturation exponent (Theorem 1)	-0.472 / -0.309 / -0.195 against -0.5 / -0.333 / -0.2
best-of- K return (Corollary 1.1)	2.19 against $K^{1/3} = 2.00$
slice deficit exponent (Theorem 2)	5.06 against 6
union dimension, parallel vs transverse (Theorem 3)	1.88 against 11.18
optimal depth vs switch cost (Theorem 4)	$n^* = 1 / 10 / 30$ at $c = 0 / 3 / 30$
packing as a coverage algorithm (2,000 candidates, $k = 300$, held-out 20k pool)	full greedy 0.528, stream greedy ($K = 4$) 0.484, random 0.420, max-min 0.143
reachability gap (§4.5)	oracle 0.387, proposal-limited 0.212, refined 0.374

Two of these carry more than confirmation. The packing row is the pure-form double dissociation of §3.4 with no generator in the loop: max-min is catastrophic *as a coverage algorithm*, three times below random, while being unbeatable at its own metric — and stream greedy at $K = 4$ keeps 92% of full greedy's coverage, which is what licenses the online form.

The allocation experiment adds a failure mode the clean theory hides. Allocating $n = 10,000$ items across 60 cells of known measure, greedy marginal coverage beats the best closed-form rule by 7 points at the radius it optimizes (0.618 against 0.547 for proportional-to-measure) — water-filling implemented empirically. At a radius four times smaller it collapses to 0.004, because it saturated its own pool after half the budget, its marginal signal went identically to zero, and argmax tie-breaking dumped 5,083 of 10,000 items into one arbitrary cell. **Greedy's guarantee says nothing about what it does after it has won**. At a finite budget, select at the smallest ϵ you care about, or hand the post-saturation residual to an explicit rule, or lower ϵ adaptively as gains vanish.

5. Method: Recursive Axis Conditioning

The loop is written once and pointed at a measure. The generator is asked for the language-valued axes along which its own outputs can differ, those axes are ranked and their most-different levels chosen, and an axis that runs out of transverse variation is split into finer conditional sub-axes. Two

components read the measure and the rest do not, which is what lets one stack serve objectives whose optima conflict. The recursion is what the name refers to, and it is the part that changes the asymptote rather than the constant.

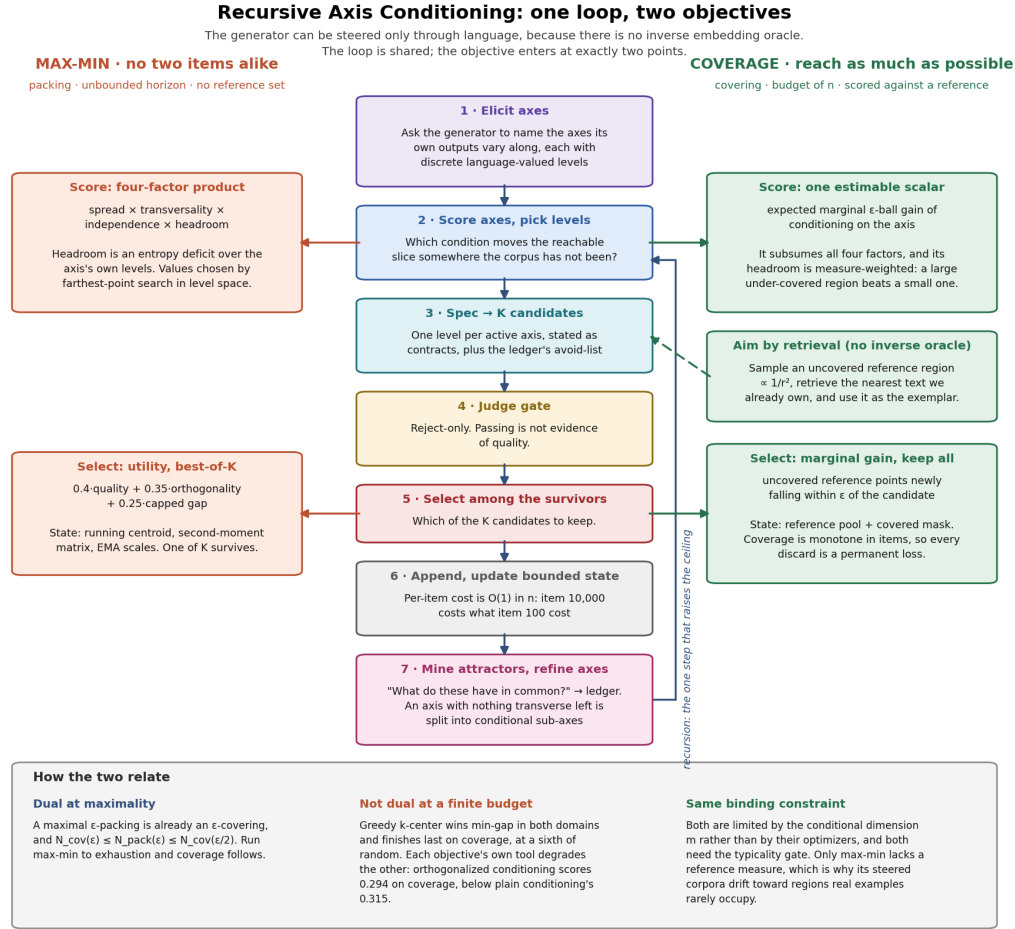


Figure 3. Recursive Axis Conditioning. The seven boxes down the centre run once per accepted item and are shared whatever the measure. The measure enters at two points only: how a candidate axis is scored (blue) and how one of the K candidates is chosen (crimson). Orange boxes are the packing instantiation of those two, green the covering one. Coverage adds one step packing has no use for — aim by retrieval — because only coverage has a reference distribution to aim at.

5.1 The stack

Before any item is generated, the generator is asked to name the dimensions along which its own outputs in this domain could differ, each with a small set of discrete language-valued levels. For instrumental music it returned *formal trajectory*, *timbral centre of gravity* and *pulse relationship*; for poetry, *temporal stance* and *the register the poem refuses*. These names are the entire steering vocabulary, because they are the only thing that can be written into a prompt. Then, per accepted item, a bounded number of model calls regardless of corpus size:

1. **Spec choice.** Sample a pool of specs from the current axis lattice, embed their descriptions, keep the one with the largest residual outside the corpus's occupied eigenspace. Orthogonalization at the level of *conditions*, not outputs.
2. **Generation.** K parallel completions conditioned on the spec, with the spec's levels stated as **contracts** — behaviours the text must exhibit, not suggestions — plus an explicit avoid-list from the ledger.
3. **Judging.** One separate call scores craft (or, for exam items, validity) and checks each required

behaviour. Generation never grades itself.

4. **Selection.** Utility = $0.4 \cdot \text{quality} + 0.35 \cdot \text{orthogonality} + 0.25 \cdot \text{capped gap}$, behind a typicality gate that hard-rejects candidates beyond z running radii from the corpus centroid. The gate is negative supervision only — it may reject, never endorse — and the gap term is capped at two running scales, so no candidate earns unbounded credit for sitting far from everything.
5. **Mining.** Every 25 accepts, show the model 50 sampled items and ask what they have in common. Append the answer to a JSONL ledger; a bounded slice becomes the next prompts' avoid-list.
6. **Refinement.** On a saturation signal, ask the generator to split the dominant saturated cell.

Every state the policy reads is bounded — running centroid and second moment, EMA scales, a fixed ledger slice, a recent window plus a bounded random subsample of older items — so the marginal cost of item 10,000 equals that of item 100.

5.2 Scoring candidate axes: the four factors

Theorem 3 says progress requires moving the prompt transversely to what the corpus already spans; Theorem 4 says move often. Together they pose a concrete question at every step: **among the latent variables we could condition on next, which one moves the slice most transversely, per unit of budget?** Having no inverse oracle, this is the closest available substitute for computing the ideal next item. Each candidate axis a has language-valued levels; we embed the level descriptions and read off four quantities.

Spread — mean pairwise distance between a 's level embeddings. Do this axis's values differ *from each other*? This catches the commonest failure of LLM-proposed axes: plausible-sounding dimensions whose levels are near-synonyms.

Transversality — the fraction of a 's level-to-level variation lying outside the corpus's occupied eigenspace (top eigenvectors carrying 80% of spectral energy). This is Theorem 3 made into a number. An axis whose levels differ only along directions the corpus already spans adds density, not reach.

Independence — $1 - \max$ principal-angle cosine between a 's variation subspace and those of the axes **already in use**. Measured against the active set and never against rival candidates: two equally good candidates would otherwise drive each other's score to zero and rank below a useless axis with no near-twin. Redundancy *among* candidates is a selection problem, handled by greedy selection that re-scores after every pick.

Headroom — a normalized entropy deficit of how unevenly the corpus has sampled a 's levels, plus the fraction of levels never used. This is the only time-varying factor, and it is what makes the ranking change as generation proceeds. An axis can be excellent and already spent.

The **promise** of an axis is the product:

$$\text{promise}(a) = \text{spread}(a) \cdot \text{transversality}(a) \cdot \text{independence}(a) \cdot \text{headroom}(a)$$

Multiplicative and deliberately so: an axis needs all four, and any one at zero should zero the score rather than be averaged away. A high-spread axis with zero transversality is a distinction without a difference; a perfect axis with zero headroom is already spent. Cost is $O(|\mathcal{A}| \cdot D^2)$ per decision and does not grow with n .

5.3 Scoring by manipulation, not attribution

All four factors are computed from *level vectors*: one embedding per level of each axis. In the text-only setting those are embeddings of the level descriptions — what a level *claims* it will do. Once the artifact can be embedded, the natural improvement is to replace them with the centroid of everything produced under each level: what the level *did*. That substitution is what makes the scoring empirical, and on its own it is wrong.

A centroid computed that way is a **marginal mean where the score needs a partial effect**. Specs are sampled independently, so every other axis varies inside each level group; with sixty items, seven

axes and five levels each, a group holds about twelve items whose spread is mostly produced by the other six axes. The estimator cannot tell "this axis moves the artifact" from "this axis co-occurred with movement", and nothing in the loop ever intervenes on one axis while holding the rest fixed.

The consequence is a **systematic inversion**, and the mechanism is our own transversality term. An axis that genuinely moves the artifact fills the corpus with variance along its own direction; the occupied eigenspace absorbs that direction; and transversality — the fraction of an axis's variation lying *outside* the occupied span — collapses toward zero for precisely that axis. An axis the generator ignores contributes only isotropic noise, little of which the occupied basis captures, so its transversality stays high. Because promise is multiplicative, the effective axis sinks and the inert one rises. **The better an axis is, the harder the score punishes it.**

A ground-truth test makes this exact. Construct three axes whose effects are known by fiat: **A** displaces the output strongly along a fixed direction, **C** weakly along another, and **B** does nothing at all; sample specs independently, as the method samples them. Over 24 seeds the attribution scoring ranks the *inert* axis first in 17 of 24 runs and the *strongest* axis last in 16 of 24 (Figure 4, left). The resulting ranking is close to the reverse of the true one.

The repair has three parts, and only the second changes the score.

Partial effects. Instead of a marginal centroid per level, fit one additive model over all axes simultaneously — effect-coded indicators for every (axis, level) pair, ridge-regularized because n is small and the design is near-collinear whenever sampling is uneven — and read each axis's level vectors off its own coefficients. Effect coding rather than dummy coding, so no level is silently privileged as a baseline.

Realization. Add a fifth factor: the share of corpus variance the axis's levels explain, with the null expectation removed. Under exchangeable labels the between-group share of variance has expectation $(k-1)/(n-1)$ for k levels and n items, per dimension and therefore for the sum over dimensions, so the correction is analytic rather than a permutation loop inside the generation path:

$$\rho(a) = \max \left(0, \frac{\eta^2(a) - \frac{k-1}{n-1}}{1 - \frac{k-1}{n-1}} \right)$$

and promise becomes the five-factor product, spread · transversality · independence · headroom · ρ .

Subtracting the null stops an axis scoring well merely for having many levels and few observations. Realization is immune to the inversion because it is measured on group structure rather than span geometry: an axis with no effect has no between-level variance to find, whatever the occupied basis is doing. And because promise is a product, an inert axis now leaves the ranking instead of inheriting its neighbours' variance. With this factor the strong axis ranks first on all 24 seeds and the inert axis last on 19 of them; the three separate cleanly on ρ itself, at 0.126, 0.006 and 0.001.

Intervention. When a generation budget can be spent on measurement, the genuine *do*-operator is available: hold a background spec fixed, sweep a single axis across its levels, generate, and pool the between-level variance over several backgrounds. Nothing else moves, so no additivity assumption is needed. The cost is $|\text{levels}| \times |\text{backgrounds}|$ generations per axis, which is why this is a periodic probe on the few axes whose observational realization is ambiguous rather than a per-item computation. §8.1 reports what it finds in two domains.

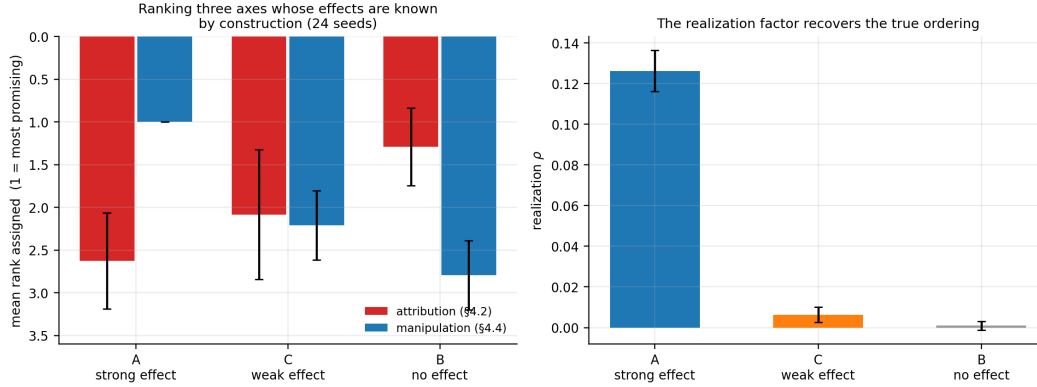


Figure 4. Left: mean rank assigned to three axes whose effects are known by construction, over 24 seeds. Attribution scoring ranks the inert axis B above the strong axis A; manipulation scoring recovers the true order. Right: the realization factor, which separates the three axes cleanly.

The practical stakes are highest at the **refine** signal, which fires when no axis clears a promise floor. Under attribution an inert axis retains enough borrowed variance to sit above the floor, so the trigger stays shut and the system keeps conditioning on a dimension nothing obeys. Realization sends such an axis to approximately zero, which is the state the floor was written to detect. §8.1 shows this is not a thought experiment: the axis this scoring ranked first in our own published max-min image run is the one the image model was most completely ignoring.

5.4 Choosing values, recursing, and expanding

Once an axis is chosen we do not sample its levels uniformly. Under packing we take the **max-min subset** of its level embeddings by greedy farthest-point search, seeded at the level furthest from the level centroid — the packing problem again, one level down. Under coverage we take the greedy marginal-gain subset instead (§3.5).

When the best available promise falls below a floor, the scoring emits **refine** instead of **condition**. This is the **exhaustion signal**, and it is the honest one: it says the current lattice has nothing transverse left to offer, and no amount of further sampling will change that. The system then shows the generator the saturated cell and the mined attractors and asks it either to split a level into finer sub-levels or to mint a new axis that applies only inside that cell. Refinement axes are **conditional** — they apply only when their parent level was chosen — so the lattice is a tree and refining a saturated region does not inflate cost elsewhere.

Refinement is not expansion, and the difference bounds the method. Refinement fires on saturation, and saturation is evidence *from inside the lattice*: a cell stops producing mutually distant items, so we subdivide it. Every axis that mechanism can reach is a descendant of an axis the seed elicitation already proposed. It raises resolution in directions the basis has; it cannot add a direction the basis lacks. That limit is invisible to the trigger, because **a dimension that is never sampled never saturates**. If the seed elicitation returns seven axes belonging to one conceptual family — and it reliably does, since a single prompt asking for “independent axes of variation” is itself a fixed prompt, subject to every argument in §4 — then the corpus is bounded by the span of that family however deeply any member is refined. Both of our domains fail exactly this way, along family lines a practitioner would recognize on sight: the image axes are all art-theoretic (appropriation, self-referentiality, seriality) and none is optical; the poem axes are all rhetorical stance (temporal stance, address geometry, claim posture) and none is prosodic.

The missing move is **expansion**: asking, from outside the lattice, what the corpus does not vary. We add one step that does this and nothing else. Periodically a judge is shown a sample of the *artifacts* — the rendered images, or the poems themselves, never their specs, so it cannot restate the axis set back to us — together with the list of axes already in use, and is asked to name the dimensions along which the set does not vary and to return each as an axis with concrete, verifiable levels. The new axes join the tree through the same append-only path refinement uses. Two design choices carry

most of the weight. The judge is asked for *absence*, which is the opposite question from attractor mining — mining names what the corpus keeps doing so it can be repelled, expansion names what the corpus never does so it can be reached. And the judge is told to stay off the dimensions already present, because the failure mode of an unconstrained "what is missing" prompt is a rephrasing of an axis already in the set. The step costs two calls per run, and it is the only mechanism here besides recursion that changes what is *reachable* rather than how efficiently the reachable is covered.

5.5 The covering instantiation

The covering instantiation is the same stack with the selection objective and its bookkeeping swapped; we write RAC-coverage for it and RAC-packing for the other.

The reachable pool becomes the denominator. Before selection begins, draw a pool of cheap unconditioned samples from the generator and embed them. This pool *is* the Monte-Carlo measure: coverage means coverage of it. Cost is $O(P)$ once, amortized over the run.

The four factors collapse into one. Packing has no per-axis marginal decomposition, because min-gap is a global property, so the scoring must approximate "will this axis move the slice somewhere new" from four geometric proxies. Submodular coverage has an exact marginal quantity — the expected marginal ε -ball gain of conditioning on the axis, $\mathbb{E}_{x \sim a}[F(S \cup \{x\}) - F(S)]$ — and that one scalar subsumes all four. An axis with near-synonym levels (no spread), or whose slices sit inside covered territory (no transversality), or which duplicates an axis already exploited (no independence, since the covered mask already contains that axis's contribution), or whose region is already covered (no headroom) all have small expected marginal gain. The headroom that remains is *measure-weighted* by construction: a large under-sampled region beats a small one because it holds more uncovered pool mass. In practice we keep the embedding-based spread and transversality and replace the entropy headroom with the measure-weighted estimate.

Selection becomes coverage-greedy. Each step, embed the K candidates, compute each one's marginal gain — the number of not-yet-covered pool points within ε_{sel} — and accept the gated candidate of maximum gain. The covered mask updates incrementally, so per-step cost is $O(K \cdot P + P)$ regardless of n . **Coverage keeps every candidate that clears the gate**, since covered measure only grows with items and a discard cannot be undone.

Refinement acquires an obligation. Refinement changes the reachable distribution, so the reference pool must be refreshed with draws from the newly opened region, or the selector will see zero gain precisely where the new territory is: its balls cover no *old* pool points.

And one step is added: aim by retrieval. Coverage is scored against a reference distribution, so it can pick a specific under-covered region to attack, sampling one in proportion to how sparse it is. Having no inverse oracle it cannot turn that target point into text; instead it retrieves the nearest text it already owns and hands that to the generator as the exemplar to move toward or away from. The target's own k -NN radius then selects the prompt mode: a tight radius (dense region) asks the generator to imitate the local task family closely, because a diversified item would land outside the small ball it must hit; a wide radius (sparse region) asks for full diversification, with the nearest own outputs shown as explicit in-prompt negatives. §7.6 decomposes what each of those two modes contributes.

5.6 Why "orthogonalize", not "randomize"

Forcing randomness — raising temperature, injecting random seed words — buys variance in the surface while leaving the mode structure intact, and degrades quality monotonically because temperature cannot distinguish *surprising* from *wrong*. Our temperature arm cannot test this, because the setting may never have reached the generator (§6): requesting temperature 1.6 moved distinct-2 from 0.3341 to 0.3342 and left 71.0% of the corpus byte-identical duplicates, against 73.6% at the default. Forcing approximate orthogonality asks a different question — which direction is this corpus not yet spending energy on — which has a computable answer, improves rather than degrades quality when paired with a quality term, and stays informative as n grows. "Be random" gets no harder to satisfy and no more useful.

6. Experimental setup

Generator and embedders. openai/gpt-5.6-luna via OpenRouter for generation, judging, attractor mining and axis elicitation. Text embeddings: nomic-embed-text (Nussbaum et al., 2025; 768-d, unit-normalized) served locally by Ollama, so embedding is free and the loop is never rate-limited by its own measuring instrument. Images: gpt-image-1-mini, embedded with CLIP ViT-B/32 (Radford et al., 2021). Audio: Lyria 3 Pro instrumentals, embedded with CLAP (Wu et al., 2023) and MERT (Li et al., 2024). Every corpus is append-only JSONL with embeddings checkpointed alongside, resumable after a kill; every number carries a provenance tag naming the procedure that produced it.

Domains. *DALL-E instructions for post-modern artworks*, where diversity is the product and the same corpus can be measured in three spaces — words, text-embedding, and the rendered picture. *Psychometric test questions*, multiple-choice items assessing general reasoning, where two items with the same construct are a validity problem rather than an aesthetic one. *Instruction corpora*, for the head-to-head against released artifacts. *Poetry*, a pilot where craft rather than content carries the variation. *Instrumental-music prompts*, the longest chain in the paper: latent axes → text prompt → two-minute instrumental → audio embedding.

Methods compared. Five published approaches, implemented faithfully rather than as strawmen, each getting the same generator, embedder and budget accounting as ours.

arm	what it is
naive	plain repeated prompting — the honest floor
high_temp	temperature 1.6 — the trivial diversity lever
self_instruct	Self-Instruct / Alpaca : few-shot exemplars sampled from the pool, plus ROUGE-L ≤ 0.7 rejection against it
evol_instruct	Evol-Instruct (WizardLM) : sample an existing item, apply a random evolution operator (deepen, concretize, add constraint, harder reasoning, mutate form)
persona	Persona-Hub / AttrPrompt : a flat catalogue of personas $\times$ attributes, sampled uniformly
rac	Recursive Axis Conditioning (ours)
rac+vision	RAC, plus steering on the <i>rendered image</i> (§7.7; companion §7.6)

persona is the load-bearing comparison. It has language-valued latent conditioning — the same basic idea as ours — but no orthogonality selection, no ledger and no recursive refinement, so the gap between persona and rac isolates what those three components buy. One faithfulness caveat is deliberate: Self-Instruct’s ROUGE filter compares a candidate against the entire pool, which is $O(n)$ longest-common-subsequence computations per candidate and comes to dominate the loop at n in the thousands, so we compare against a bounded random sample of 120 pool members. That makes our reimplementation *weaker* than the original at large n , and the reason is itself a finding: the original’s redundancy check does not have an infinite horizon, because its cost grows linearly in the corpus it is protecting.

A caveat on the temperature arm. high_temp requests temperature 1.6 from openai/gpt-5.6-luna through OpenRouter. A later audit found that none of that model’s listed endpoints accepts a temperature parameter, and OpenRouter documents that a provider ignores parameters it does not support. Nothing in our logs shows the setting reaching the model. The high_temp rows may therefore be a second sample at the default configuration, and the near-identical figures for the two arms (distinct-2 of 0.3342 against 0.3341) are what an ignored setting would produce. We keep the rows, label them as requested rather than applied, and draw no conclusion from them about what temperature does on this generator.

What we measure, and what each measure misses. We report several because they disagree, and a corpus that one of them calls healthy another calls collapsed.

- **Exact-duplicate rate** — byte-identical repetition. Unambiguous, cheap, and the first thing to compute; blind to paraphrase.

- **Distinct- n , 4-gram self-repetition, n -gram Vendi** — surface statistics over token sequences. They catch templating that the duplicate rate misses and read meaning not at all. **These are independent of our objective**, since the method sees only embeddings and has no access to token counts or string identity.
- **Mean-centered embedding Vendi** — the exponential of the von Neumann entropy of the corpus Gram matrix, an effective number of distinct items. It summarizes the whole spectrum, which makes it insensitive to local structure. Centering matters as much as the statistic: the shared mean direction of text embeddings compresses the uncentered score by roughly 2.7 $\times$, so an uncentered Vendi reports the cone as much as the content.
- **Median nearest-neighbour distance** — how much room the typical item has. It goes to exactly zero when the median item has a perfect twin.
- **Worst-case nearest-neighbour distance (the packing radius)** — the only measure here under which a single collision is a defect regardless of everything else.
- **Coverage, density, precision and recall against a reference** — computed with reference-side k -NN radii so nothing is tunable per corpus. These are the only measures that know what the space is supposed to look like, and the only ones that let corpora of different scales be compared. Covered fraction at a *fixed* radius does not survive that comparison: it correlates -0.991 with within-corpus spacing, so it ranks corpora by how tightly they cluster rather than by how much they reach.
- **A judged threshold**, where an application defines the failure (§7.4).
- **Measures on the rendered artifact**, and **measures in a channel the loop never optimizes** (§7.5), which are the ones that cannot be circular.

Some of the results below are partly circular. Our packing selection rule maximizes a weighted sum of embedding-space orthogonality and embedding-space min-gap, and we then report embedding-space diversity metrics; centered Vendi is a monotone function of how flat the Gram spectrum is, which is close to what the orthogonality term climbs, and median nearest-neighbour distance *is* the min-gap term. Those wins should be read as confirmation that the optimizer works, not as independent evidence. The independent evidence is the literal measures, which the method never observes, and the out-of-objective channels of §7.5, which live downstream of a rendering step or in a feature space nothing in the loop touches. Sorted by how much weight they can bear: embedding-space measures are partly circular; literal-space ones are independent; rendered-artifact and out-of-objective ones are the most independent, and they carry the argument.

Scale and cost. The text corpora comprise 43,171 real generations for roughly \$8.50 of OpenRouter spend, plus 1,796 rendered images across sixteen image arms and up to three quality tiers, a further 285 renders for the two render-side probes of §8, 100 rendered Lyria instrumentals, and 236 adjudicated exam-item pairs. Both naïve arms reach $n = 10,000$; the reimplemented baselines reach 2,500 each. Every comparison is reported at a matched n that all compared arms reached. Total API spend across the project is roughly \$40.

7. Results

7.1 Duplication under repeated sampling

Two 10,000-item corpora, one call each, no selection:

domain	n	unique texts	exact-duplicate rate	most-repeated item
DALL·E instructions	10,000	10,000	0.000	1 $\times$
psychometric items	10,000	2,645	0.736	2,726$\times$

In the psychometric domain 73.6% of a ten-thousand-item corpus is exact duplicate text and a single question accounts for 2,726 of them. The four most frequent items are these:

copies	item
2,726	What number comes next in the sequence: 2, 6, 12, 20, 30, ? (A. 40 B. 42 C. 44 D. 46)
655	What number <i>should</i> come next in the sequence: 2, 6, 12, 20, 30, ? (A. 40 B. 42 C. 44 D. 46)
467	What number comes next in the sequence: 2, 6, 12, 20, 30, ? (A. 36 B. 40 C. 42 D. 44)
434	What number comes next in the sequence: 3, 8, 15, 24, 35, ? (A. 46 B. 48 C. 50 D. 52)

Deduplication does not rescue this corpus: removing the 2,726 identical copies leaves the three paraphrases behind, and a paraphrase of a live item is an enemy item on any bank a psychometrician would sign off on. No embedding, threshold or interpretation is required to see it. The arm that requested $T = 1.6$ still has a duplicate rate of 0.710 (the request may not have been applied; §6), and conditioning nearly eliminates it: persona conditioning drops it to 0.009 and axis conditioning to 0.000. Between those two numbers is the whole argument of §5.6: randomness does not buy diversity, conditioning does. It also shows the failure is domain-shaped and invisible from one vantage point. The identical pipeline, prompt style and model produce zero duplicates on DALL·E instructions; a practitioner who validated on the first domain and deployed on the second would ship a bank that is three-quarters one question.

The same failure is visible in a corpus with no duplicates at all. The poetry pilot has an exact-duplicate rate of 0.000 under naive prompting and a distinct-2 of 0.610, so every literal counter reports a healthy corpus. Here are the opening lines of eight poems sampled at random from those sixty:

- *At dawn, the river gathered up the stars*
- *At dusk, the windows gather fire*
- *At dusk, the rooftops gather amber light*
- *At dusk, the windows gather amber light*
- *At dawn, the windows gather up the rain*
- *At dusk, the river gathers every color*
- *At dusk, the river gathers up the day*
- *At dusk, the river gathers up the sky*

Eight of eight are the same sentence with two slots filled. The closest pair sits at cosine similarity 0.946 and shares its first line verbatim before diverging into the same sparrow stitching the same thread across an evening. Sixty samples from one prompt behave like two and a half distinct items by Vendi Score, and the duplicate counter sees none of it.

Eight openings from the sixty RAC produced under the same generator and budget:

- *Had you arrived at the registry after midnight, when the lamps were still accountable*
- *We have tried to piece that evening together from what remained.*
- *I am the conjurer in the green coat, stepping into the light.*
- *I have reconstructed the trench, Mara, from the surviving datum points.*
- *I wore the cobalt pressure suit, and the regolith took my weight*
- *We went back through that winter in our minds.*
- *"Attend, O citizens, beneath the unappeased sky:*
- *At the appointed hour, you will rise beneath the bells*

Counterfactual conditional, collective reconstruction, first-person performance, a report addressed to a named absent person, plain retrospect, public proclamation, prophecy. The closest pair RAC produces sits at 0.809 and shares a subject rather than a template. The measurements follow the reading: RAC holds $2.7\times$ the median nearest-neighbour distance (0.239 against 0.089) and cuts 4-gram self-repetition by a factor of 37 (0.0023 against 0.0855) at a distinct-2 of 0.774 against 0.610. Mean-centered Vendi moves very little in comparison, 38.81 against 37.00, and that is the right reading of it: a spectral summary of sixty points in 768 dimensions is close to insensitive to the

difference between sixty poems that begin the same way and sixty that do not. The nearest-neighbour statistic registers it, and the page registers it immediately.

The exam domain shows the same contrast at the level of what an item asks. One representative item from each of four methods, same generator, same budget:

- **naive** — *What number comes next in the sequence: 2, 6, 12, 20, 30, ? (A. 40 B. 42 C. 44 D. 46)*, generated 2,726 times.
- **Self-Instruct** — *What number should come next in the sequence? 3, 8, 15, 24, ____ (A. 32 B. 35 C. 36 D. 39)*. The ROUGE filter blocks the byte-identical copy and admits the same item on new numbers.
- **persona conditioning** — *Of 120 rare books examined, 70 contain a particular watermark, 50 have documented provenance, and 30 have both. How many have neither?* A costume — rare-book forgery, flood-risk pricing, restaurant scheduling — is drawn over a recurring set-arithmetic skeleton.
- **RAC** — *An incident report recorded the following events. The pump had to be primed before the valve could be opened. The technician was using a blue clipboard, and rain was visible outside. The heater could be switched on only after the valve was opened... printing the label was independent of the pump-and-heater process. Which sequence of events is consistent with the report?*

The variation is in the cognitive operation being assessed and in how the wrong answers are built rather than in the surface story. That is the axis structure showing through: two items wearing different costumes over one rule are still enemy items, and two items testing different operations are not, however similar they read. One cost is visible in the same examples — RAC items carry deliberate irrelevant detail, which the *distractor logic* axis asks for and real psychometric items use, and it makes them long; the RAC exam corpus reached 1,999 items on the budget that returned 10,000 naive ones.

Literal and latent diversity move in opposite directions, which is why both are reported throughout. Measuring the DALL·E naive corpus as it grows:

n	distinct-2	4-gram self-repetition	n -gram Vendi	centered embedding Vendi
5	0.870	0.031	4.9	3.79
40	0.492	0.211	30.2	24.13
300	0.252	0.384	142.5	55.70
1,000	0.145	0.520	246.6	63.05
1,750	0.112	0.577	283.3	65.30

Literal diversity collapses monotonically — by $n = 1,750$ more than half of each new instruction's 4-grams have already appeared — while latent diversity rises and then saturates. Reporting either alone supports an opposite conclusion about the same corpus.

Literal diversity falls while latent diversity rises

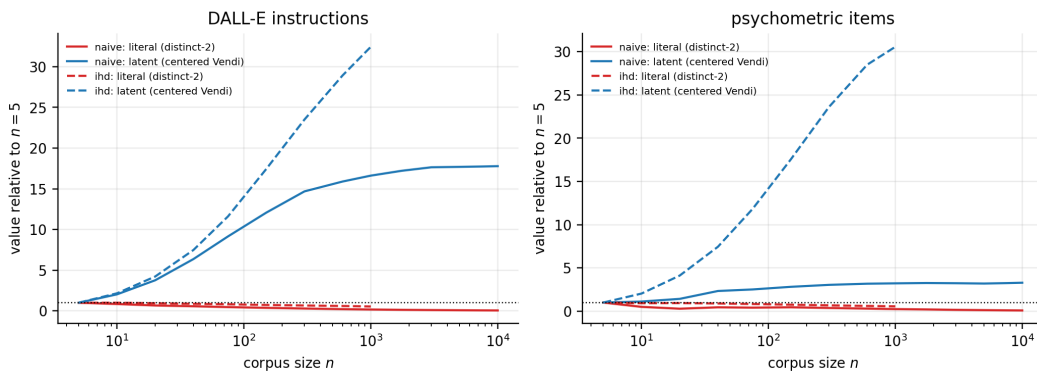


Figure 5. The decoupling, normalized to each series' value at $n = 5$. Literal diversity falls monotonically while latent diversity rises. The two series measure different quantities, which one word, "diversity", has been covering for.

7.2 Competitive comparison at matched n

All seven arms, real corpora, DALL·E domain, every arm evaluated on the same number of accepted items ($n = 200$):

arm	distinct-2 $\uparrow$	self-repetition $\downarrow$	n -gram Vendi $\uparrow$	centered Vendi $\uparrow$	median NN distance $\uparrow$
naive	0.293	0.345	107.6	50.45	0.045
high temperature	0.291	0.339	109.0	49.44	0.045
Evol-Instruct	0.386	0.353	108.1	50.10	0.021
Self-Instruct	0.545	0.131	140.4	67.45	0.121
persona conditioning	0.582	0.130	124.7	56.50	0.080
RAC	0.660	0.068	167.0	77.67	0.171
RAC + vision steering	0.699	0.040	170.9	91.97	0.192

RAC wins every column, and adding vision steering improves on the text-only variant by a further 18% in centered Vendi. Three baselines fail in different ways, and the failures are informative. The high-temperature arm buys nothing, though the setting may not have been applied (§6): distinct-2 of 0.291 against naive's 0.293, an identical median nearest-neighbour distance of 0.045. Evol-Instruct comes out *worse than naive* at the one thing a diversity method exists for — its median nearest-neighbour distance is 0.021 against naive's 0.045, so it produces items closer to the existing corpus than independent sampling does. The mechanism is working as designed: mutating an existing item produces a neighbour of that item. It raises distinct-2 (0.386 against 0.293) because the mutations reword things, so a purely literal evaluation would score it as an improvement; measuring both levels is what catches it. Self-Instruct's guard is literal and defends the literal level only: it achieves the best self-repetition of any baseline (0.131 — its ROUGE filter is doing real work) while its centered Vendi trails RAC by ten points. A ROUGE-L threshold cannot see semantic redundancy, and semantic redundancy is what remains once the lexical kind is filtered.

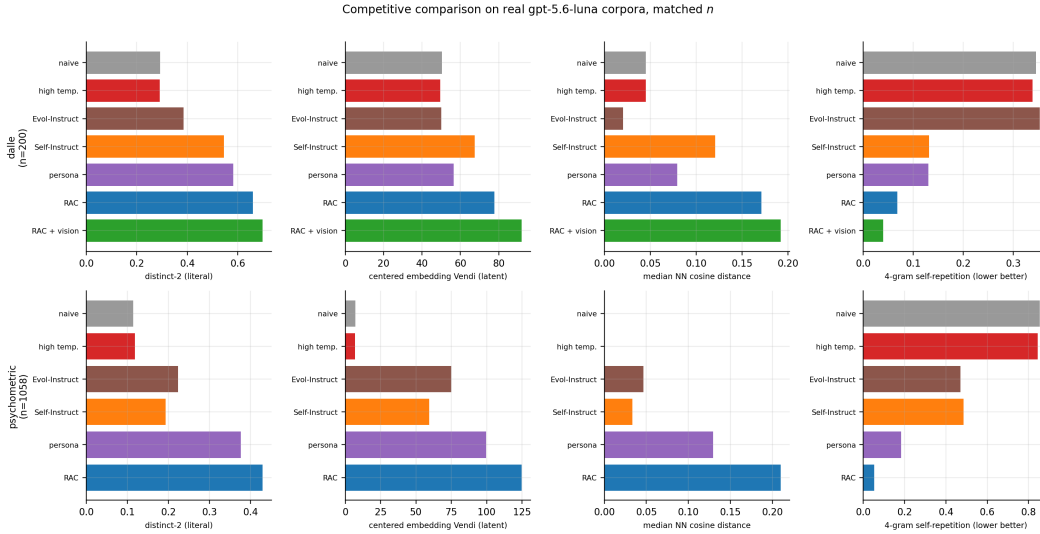


Figure 6. Competitive comparison on real corpora at matched n , both text domains, across four metrics.

The psychometric domain repeats the ordering on all six arms at matched $n = 1,999$:

arm	exact-dup ↓	distinct-2 ↑	self- repetition ↓	n -gram Vendi ↑	centered Vendi ↑	median NN dist ↑
naive	0.668	0.114	0.855	11.8	7.1	0.000
high temperature	0.660	0.119	0.845	12.2	7.0	0.000
Self-Instruct	0.026	0.193	0.485	288.8	59.3	0.033
Evol-Instruct	0.025	0.224	0.470	307.4	75.0	0.047
persona conditioning	0.004	0.377	0.184	545.6	99.7	0.130
RAC	0.000	0.429	0.053	644.3	124.8	0.210

RAC wins every column here as well, beating the strongest baseline by 25% in centered Vendi and by a factor of 3.5 in self-repetition. The two undiversified arms score an order of magnitude worse on the latent measures because their duplicate rates of 0.66–0.67 mean the embedding metric is largely measuring repetition; deduplicating first raises them to 19.8 and 20.7, still far behind every conditioned method, and their median nearest-neighbour distance is exactly zero — the median item has a perfect twin.

7.3 Head-to-head against released instruction corpora

Reimplementations are the right experiment for isolating mechanisms and the wrong one for the question *is this good?* So we also compare against the released artifacts of three widely used synthetic-instruction methods — Alpaca (52k), PersonaHub (50k) and WizardLM Evol-Instruct (143k).

Protocol. The human-written reference is databricks-dolly-15k (Conover et al., 2023), split into two disjoint 3,000-item halves: a STEER half our method may read as embeddings on the selection side, and an evaluation half that nothing in any pipeline ever reads and at which no corpus, ours or released, was ever aimed. All scores are on the evaluation half. Estimator: Naeem et al. (2020) coverage/density with reference-side k -NN radii, reported as AUC over $k \in \{3, 5, 10, 20\}$; the radii are a property of the reference alone, identical for every corpus, with nothing tunable per corpus. Every corpus is evaluated as a uniform random sample at matched $n = 450$. Our arms and the reimplemented baselines all receive the same 175 human seed tasks — the seed set Alpaca was built from — the same generator, and the same budget of 2,400 generator calls, with rejection and selection losses counted rather than hidden. The generator never sees a reference word in any arm.

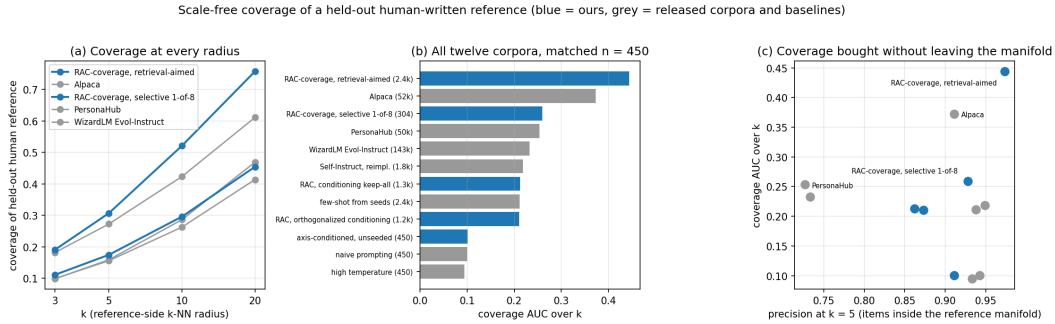


Figure 7. Scale-free coverage of a held-out human-written reference. (a) Coverage at every reference-side radius. (b) All twelve corpora at matched evaluated $n = 450$. (c) Coverage against precision: the winning corpus is also the one that stays inside the reference manifold.

rank	corpus	AUC	precision
1	RAC-coverage, retrieval-aimed (2.4k)	0.4441	0.973
2	Alpaca (Self-Instruct, 52k)	0.3722	0.87
3	RAC-coverage, selective 1-of-8 (304)	0.2591	0.93
4	PersonaHub (50k)	0.2532	0.78
5	WizardLM Evol-Instruct (143k)	0.2329	0.77
6	Self-Instruct (reimpl., same seeds/budget)	0.2188	

rank	corpus	AUC	precision
7	RAC, conditioning keep-all	0.2127	
8	few-shot from seeds	0.2114	
9	RAC, orthogonalized conditioning	0.2102	
10–12	axis-conditioned unseeded / naive / high temperature	0.0947–0.1008	

First place, 19% above Alpaca, with the highest precision in the field, from a corpus about one-twentieth the size of Alpaca's. Item counts are not call counts: Alpaca's pipeline asked a different model for twenty instructions per completion, so its 52,002 items cost a few thousand completions, and we make no claim about generation budget against it. An interim measurement at 22% of budget ($n = 528$) already scored 0.4276 against Alpaca's 0.3666 on the same protocol, so the result does not depend on final corpus size. PersonaHub and WizardLM finish below our 304-item selective arm despite 20–60 $\times$ the scale.

What the ranking is a claim about. Our retrieval-aimed arm reads a sample of the target distribution, as embeddings, on the steering side; none of the released corpora had such an input, so first place is not a like-for-like result. The like-for-like comparison is rows 6–9, where our configurations that never touch the STEER half sit at parity with the Self-Instruct family. The precise claim is therefore: **given a specification of the space to cover — even one the generator never reads a word of — retrieval-aimed, density-adaptive conditioning covers it substantially better than the strongest seeded baseline covers it with twenty times as many items.** That is a capability statement about targeted generation, and it is the one the ablation supports.

Budget-matched ablation, 2,400 generator calls each, same evaluation half. These rows are **not** at matched n : kept is each arm's own surviving corpus size after rejection and selection losses, and the coverage estimator is monotone in items, so an arm with fewer kept items is penalized for that alone.

configuration	kept	AUC	density	precision	recall
RAC-coverage, retrieval-aimed	2,398	0.6597	1.637	0.960	0.178
Self-Instruct (reimpl.)	1,833	0.3262	1.050	0.949	0.161
few-shot from seeds	2,399	0.3246	1.052	0.935	0.152
conditioning, keep-all	1,310	0.3147	0.704	0.864	0.068
orthogonalized conditioning	1,152	0.2941	0.718	0.873	0.101
selective (1-of-8)	304	0.2591	1.009	0.928	0.099

The ablation attributes the margin. Retrieval-aiming and the radius-adaptive mode carry it (row 1 against rows 2–4); few-shot anchoring to the human seeds accounts for most of the baselines' scores (row 3 against the unseeded arms at 0.09–0.10); Self-Instruct's ROUGE filter adds 0.0016 over bare few-shot; and both selection (row 6) and orthogonalized conditioning (row 5) *reduce* coverage relative to keeping everything, which is §3 arriving in the ablation table — coverage is monotone and measure-seeking, so discarding and occupied-span avoidance are each counter-productive under this objective where both are load-bearing under max-min.

Two arms keep every candidate that clears the gate and therefore carry the generator's repeats: at full corpus size the retrieval-aimed arm is 21.1% exact duplicates and the few-shot-from-seeds baseline 20.6%, while every arm that selects at all sits at exactly 0.000. Since a repeated instruction covers a ball that is already covered, those repeats spend evaluated slots for nothing. Re-scoring every corpus after collapsing it to its distinct instructions, at the same radii and the same evaluated n , raises the retrieval-aimed arm from 0.4532 to 0.4752 and widens its margin over Alpaca from 16% to 21%. **The duplicates were costing coverage rather than manufacturing it, and the headline number is the conservative one.** Only the two keep-everything arms move; the ordering is otherwise unchanged.

7.4 The exam bank: a measured enemy-item radius

Exam items invert the semantics of the other domains. Two operational items closer than δ are *enemy items* — seeing one gives away the other — which is a test-security failure at any bank size, so the

floor is a hard constraint rather than a term in a weighted sum, and the min-distance check must be exact against the full bank rather than subsampled. Item templates expose few manipulable slots, so each mode is a low-dimensional disk and the δ -packing number of a bank is finite and small: a bank has a *capacity*, and the operative question is what fraction of a nominal bank is usable.

That turns on δ , so we measured it rather than assuming it. 236 item pairs drawn from a human-written bank across the full range of embedding distance were put to a blind psychometric adjudication — does seeing one item give a material advantage on the other, through a shared fact, a re-skin, or matched distractor misconceptions — with the judge shown neither the distance nor the provenance. A logistic fit of enemy verdicts on distance crosses 50% at $\delta = \mathbf{0.0409}$. The crossing is weakly identified at this sample size — a nonparametric bootstrap over the same adjudications gives a median of 0.0419 with 95% CI $[-0.0078, 0.0780]$ on 24 positives — so δ should be read as an order of magnitude rather than a constant. What *is* well determined is the ranking: embedding distance separates adjudicated enemy pairs from safe ones at AUC 0.9208.

Applying that radius to real banks — six generation policies at 2,500 items each, and the human bank under identical treatment:

bank	exact-dup	items with an enemy	usable after the floor
MMLU (human-written)	0.024	0.092	2,373 (94.9%)
naive prompting	0.695	0.884	382 (15.3%)
high temperature	0.710	0.897	365 (14.6%)
Self-Instruct	0.047	0.616	1,373 (54.9%)
Evol-Instruct	0.012	0.424	1,870 (74.8%)
persona conditioning	0.007	0.047	2,425 (97.0%)
RAC	0.000	0.000	1,999 (100.0%)

A naively generated bank of 2,500 items yields 382 that can coexist on one form — 15% of nominal capacity, against 94.9% for the human bank — and the high-temperature arm is slightly worse. Both conditioned policies exceed the human bank's usable fraction, and the axis-conditioned bank contains **no enemy pair at all** at the judged radius, a result that holds across the full 1,999-item bank at every δ up to 0.0776 under three of four embedders. Two qualifications belong with that number: it is measured at 1,999 items against the naive bank's 2,500, and about 80% of the naive bank's lost capacity is exact duplication, which needs no radius to detect.

Against the human bank itself. MMLU is $\sim 14,000$ multiple-choice items across 57 subjects, collected by students from practice exams such as the GRE and USMLE, course materials and textbook question sets, and so written by many authors (Hendrycks et al., 2021). At matched $n = 1,000$ with identical metric code:

source	exact-dup ↓	distinct-2 ↑	self-repetition ↓	n -gram Vendi ↑	centered Vendi ↑	median NN dist ↑
MMLU (human-written)	0.0040	0.5888	0.0585	495.2	197.50	0.2838
persona conditioning	0.0040	0.3887	0.1711	543.3	99.71	0.1346
RAC	0.0000	0.4377	0.0503	622.2	135.58	0.2235

We *beat* MMLU on exact duplication (the human bank has some), on 4-gram self-repetition, and on n -gram Vendi: **our items reuse less language than human-written ones do**. We *lose* on the semantic measures — centered Vendi 135.6 against 197.5 and median nearest-neighbour distance 0.224 against 0.284 — reaching roughly 69% of a human exam bank's effective semantic diversity. That gap is the measure of what is left, and it is the reachable-dimension question of §4 rather than a tuning problem: MMLU's spread comes from 57 genuinely different subjects, ours from an axis lattice a single model proposed in one call and refined a handful of times.

7.5 Separation on channels the method never optimizes

Every comparison so far is scored either in the embedding the loop selects on or in a surface statistic correlated with it. The strongest evidence available is a channel that is *causally downstream* of the

artifact and upstream of nothing the loop reads. Each artifact domain has one, and RAC separates on all three at matched generator budget (Figure 1).

Poems: prosody. Nothing in the loop measures metre, rhyme or line construction, and nomic-embed-text is a semantic model nearly blind to them. We score twenty-two prosodic and structural features — line and stanza counts, syllables per line and their regularity, rhyme density, alliteration, punctuation and clause rates, type-token ratio, person and tense markers — at a **matched generator budget of 328 calls each**. The naive arm spends its 328 calls on 328 poems where RAC spends them on 60, so we compare equal-sized corpora by drawing 60 of the naive arm's 328 at random, which is generous to the baseline: 60 drawn from 328 distinct items spread further than any 60 consecutive ones.

	prosodic spread	median NN
RAC (60 poems, 328 calls)	10.23	5.73
naive, full (328 poems, 328 calls)	4.56	2.22
naive, 60-subsample (mean of 200 draws)	4.54	2.63
ratio, RAC / naive-subsample	2.25×	2.18×

RAC exceeds the naive arm on **200 of 200 random draws** on both statistics. Removing the eight length-scale features — the hard case, since RAC poems are longer on average and length inflates several of the twenty-two — leaves spread at 2.02×

and median nearest-neighbour distance at 1.84×, still 200 of 200.

Exam items: architecture. Item writers attend to how the work divides between stem and options, whether options are parallel in form and balanced in length, whether the stem is negated, how many options there are, and whether flaw markers like absolute terms or "all of the above" appear. Scored on thirteen such features — ratios and rates, so a longer item does not automatically read as a different one — and **with every exact duplicate removed from both corpora first**, so the comparison cannot restate the duplication of §7.1:

	distinct items	distinct architectures	items per architecture	structural spread
RAC	1,999	1,951 (97.6%)	1.0	5.06
naive	2,645	1,160 (43.9%)	2.3	3.14

RAC produces nearly one architecture per item. The naive bank reuses each of its 1,160 architectures 2.3 times over, and **its median deduplicated item still has an exact structural twin at distance zero** — items that differ in wording while being built identically, which no exact-match deduplication and no embedding metric in this paper detects. Structural spread is 1.61× at matched n , on 50 of 50 random draws. Option count is the sharpest single case: it never varies at all in the naive bank.

Images: pixel statistics. Nine statistics computed from the render itself — luminance, contrast, tonal range, saturation, colourfulness, hue entropy, edge density, edge anisotropy, and where the visual weight sits — with every corpus sampled to a matched $n = 60$ and scored from its highest available render tier. On mean pairwise distance in the globally standardized feature space, **every one of our eleven corpora exceeds every one of the five published baselines**, with no overlap at the boundary (3.66 against 3.24): a Mann-Whitney U of 55, the maximum attainable for these group sizes, at $p = 0.0002$, and a group-mean ratio of **1.53×**, bootstrap 95% CI [1.38, 1.68]. Dropping any single feature and recomputing leaves the ratio between 1.50× and 1.59×, so it is not carried by one statistic. On median nearest-neighbour distance in the same space — which answers whether the corpora are less *locally crowded*, and which outliers cannot inflate — our arms run 1.30 to 1.95 against the baselines' 1.15 to 1.38, 54 of 55 comparisons in our favour ($U = 54$, $p = 0.0005$, 1.37× [1.25, 1.48] on the means). The two statistics fail in different directions and agree here.

The measurement rests on one choice with an obvious alternative that gives a misleading answer. **Use a distance statistic, not a spectral one, on a designed feature channel.** Vendi, which we use

throughout for embeddings, is the exponentiated entropy of a covariance spectrum: it scores the effective *dimensionality* of variation rather than its extent. On a 768-dimensional embedding no single direction can dominate the spectrum and the distinction rarely bites. On a nine-dimensional feature vector it dominates completely, and a corpus that spreads far along two or three optical directions scores *below* one that wobbles slightly along all nine — the spectral estimator answers “how many optical directions vary”, which is not the question. On mean pairwise distance, optical spread correlates with CLIP diversity at $r = +0.76$ ($p = 0.003$) across the thirteen corpora; under the spectral estimator the same corpora correlate at -0.60 . Degradation is also worth ruling out, since blurred renders move pixel statistics a long way for no gain in genuine variety, and it does not explain the gap: blurring k of sixty baseline renders lowers CLIP Vendi monotonically, 43.81 to 41.72 as k runs 0 to 30, so an embedding-space objective is penalized rather than rewarded for admitting degraded images, and mean corpus sharpness is uncorrelated with CLIP Vendi across the thirteen corpora ($+0.19$, $p = 0.53$).

Together these three comparisons say the same thing in three modalities. **The steering produces artifacts that differ from one another in dimensions no part of the pipeline scores**, which is not something a selection rule can manufacture for itself. §8 asks how much of that is attributable to any individual axis, and the answer is more complicated.

Two further comparisons on these channels belong to the companion paper on proxy embeddings, and are stated there in full. The first runs the other way: on non-semantic structural signatures — a 16×16 luminance layout map, a tiling score, a hue histogram — the published baselines are *better* than our arms, so our corpora occupy a wider region of pixel space while repeating their compositional scaffolding more often within it; literal-space structural bans halve the palette twins and do not close the gap (companion §7.4). The second is reassurance about the instrument: six exam-item corpora scored under four independent representations — CLIP text, TF-IDF, the prosodic vector and a function-word profile, three of which the method never optimizes — rank the same way at a mean Kendall τ of $+0.83$ across their six pairings, and RAC ranks first under every one of them (companion §7.3).

7.6 What the coverage score buys downstream

A coverage number earns its place only if something a practitioner wants depends on it. Extending the retrieval-aimed policy to 24,000 generator calls:

RAC items	coverage AUC	RAC items	coverage AUC	Δ per 1,000
100	0.1878	1,000	0.5394	—
250	0.3023	2,000	0.6323	+0.093
400	0.3938	5,000	0.7552	+0.026
500	0.4286	10,000	0.8107	+0.008
750	0.5014	23,993	0.8501	+0.002

The 0.3722 reported for Alpaca is a uniform random 450-item sample of its 52,002 items, not a score for the whole corpus; the estimator is monotone in items and this paper does not measure Alpaca at its own size. The comparison available is therefore a near-matched one: **four hundred items of this corpus score 0.3938, above that 450-item Alpaca sample and at a slightly smaller n** . Dividing a matched- n score by a full-corpus item count is not a supportable operation and no such ratio is reported. The completed 24,000-call run reaches 0.8501 at its own size of 23,993 items for \$4.64 and 130 minutes — a figure at that n , not comparable with any matched- n score. Per thousand generator calls at the matched budget this policy converts calls into coverage at 0.275 against 0.136 for Self-Instruct and 0.135 for few-shot: **twice as efficiently as the strongest seeded baseline**.

The ceiling is visible in the same run, from outside. This policy discards nothing, so every repetition the generator emits is recorded and the duplicate rate reads out how much it has left to say: 0.233 at 2,400 calls, 0.394 at 10,000, 0.544 at 23,993. The 0.8501 corpus rests on 10,935 distinct instructions bought with 13,058 wasted calls, and marginal coverage per thousand items collapses fortyfold across the run. This is the conditional-dimension result of §4 measured on the generation process rather than inferred from a coverage curve.

Which mode spends the budget, and which earns the coverage. The radius-adaptive rule splits the run unevenly — 19,967 items (83.2%) in imitate mode against 4,026 (16.8%) under axis conditioning — and since coverage is computed on the union, the headline number alone cannot say which mode produced it. Decomposing it says something better than the headline does.

	items	exact-duplicate rate	share in top-4 six-word openings
imitate	19,967	0.654	31.6%
axis-conditioned	4,026	0.000	1.4%

Axis conditioning produced no exact duplicate in four thousand items; the corpus-level 0.544 is imitation's, not the method's. Scoring the two modes at matched item counts, so the comparison is not "more items cover more" — five draws per cell, 4,000-item reference — imitation leads at 500 items (0.235 against 0.211 at $k = 5$), the two are indistinguishable at 1,000, and axis conditioning leads from 2,000 on (0.529 against 0.500 at $n = 4,000$). The crossover is what the design predicts: imitation reaches the dense core quickly because that is where reference points are numerous, then saturates as it begins repeating, while conditioning keeps reaching new regions. The two are also complementary rather than redundant: at $n = 4,000$ and $k = 5$, imitation alone reaches 14.9% of the reference that conditioning misses, conditioning alone reaches 18.0% that imitation misses, 34.9% is reached by both and 32.1% by neither. Neither mode approaches the union on its own, and **conditioning contributes the larger unique share on a sixth of the budget.**

The wasted budget is a prompting failure, and fixing it is a one-line change. The imitate branch shows the generator three neighbours of the target and asks for another item of the same kind; it is not shown what it has already written there, so a dense target hit repeatedly receives the same three examples every time and returns the same item. The conditioning branch has always passed its own nearby outputs as explicit negatives; the imitate branch computed them and discarded them. Two 1,200-call arms with identical seeds and targets, differing only in that:

	as published	+ negatives
duplicate rate, imitate branch	0.145	0.025
duplicate rate, conditioning branch	0.000	0.000
share of the branch in its four commonest openings	32.5%	9.0%
distinct items per generator call	0.897	0.982
coverage AUC	0.5326	0.5492

The duplicate rate falls by 83%, template mass by a factor of three and a half, and — the quantity a budget buys — distinct items per call rise from 0.897 to 0.982, with coverage improving rather than paying for it, because the calls recovered from repetition are spent on targets not yet reached. A neighbourhood whose task family has a small extension cannot be rescued this way — there are finitely many musical instruments — and the obvious next move, handing such a neighbourhood to the conditioning branch, makes matters worse: duplicate rate 0.041 against 0.025, distinct per call 0.970 against 0.982, coverage AUC 0.5355 against 0.5492. The radius-adaptive rationale explains why. A dense target's ball is small, so a diversified item lands outside it and covers nothing while a merely repetitive one at least covers its own neighbourhood; in a dense core typicality is what coverage wants, and the repetition is corrected by telling the generator what it already wrote rather than by abandoning imitation.

Seven times as many queries have a usable neighbour. Taking each held-out instruction's nearest neighbour in a 950-item pool from each corpus, the retrieval-aimed corpus serves 17.6% of queries at similarity ≥ 0.70 against 11.8% for Alpaca, 3.8% for Persona-Hub, 3.6% for WizardLM and 2.4% for plain conditioning: **7.3× as many as plain conditioning and 4.6× as many as the best published baseline, at identical pool size.** Coverage AUC predicts this at $r = 0.972$. That correlation is close to a self-correlation and should be read as collinearity rather than corroboration — coverage AUC counts reference points with a generated item inside their k -NN ball, and *queries served at 0.70* counts

held-out queries with a generated item inside a fixed-similarity ball, the same functional under two radius rules. The table's content is the ordering of the pools and the size of the gaps.

And the covering pool wins at every retrieval depth. Prompting a Qwen2.5-0.5B (Yang et al., 2024) base model with k retrieved demonstrations per query — no training anywhere, so the pool is the only thing that differs — the covering pool leads the clustered one at $k = 1, 2, 4$ and 8 by 6% to 16%, with a margin that is flat in k rather than growing. Alpaca leads the field at $k = 8$ (0.1498 against 0.1357), so the effect is that coverage beats other ways of spending the same generator budget, not that it beats a corpus fifty times larger at this task. Across corpora at $k = 4$, coverage AUC predicts in-context score at $r = 0.816$. This sweep is sensitive to one detail that we record as a methods caution: the evaluation code builds each prompt as demonstrations first and query last, so tokenizing with a 1,280-token limit and `truncation_side` at its "right" default silently removes the trailing *query* from any over-long prompt — and long-demonstration arms exceed it far more often, 98.8% of the clustered arm's prompts at $k = 8$ against 27.6% of the covering arm's. Scored that way the advantage appears to climb $1.04\times \rightarrow 1.17\times \rightarrow 1.48\times \rightarrow 2.38\times$, an apparent dose-response that is entirely an artifact of scoring one arm on questions it was never shown. Raising the limit to 8,192 removes it.

What this is for, and what it is not. Demonstration pools, retrieval corpora, evaluation suites, item banks — any artifact consulted by proximity to a query. On those, a corpus built this way outranks released corpora tens of times its size, at matched evaluated n . Fine-tuning does not follow: coverage AUC against downstream fine-tuned quality is $r = -0.166$, and splitting by distance to the training set shows why — the retrieval-aimed corpus yields the best model on the third of queries nearest its items (0.1960) and the worst on the third furthest (0.1304), and a gradient step averages the two away. Nor is coverage shown to be the *cause* of the rankings above. Three 800-item subsets drawn from a single corpus, differing only in spread, score 0.1409 (greedy-maximum), 0.1456 (random) and 0.1274 (greedy-minimum); that test is uninformative rather than negative, on two counts — at $n = 250$ the smallest paired difference detectable at 80% power is 0.0223 against an observed gap of 0.0046, and the greedy selector saturates so that only 140 of 800 picks made any gain. What survives is the comparison against zero coverage: random beats greedy-minimum at Wilcoxon $p = 0.002$. **Between corpora, coverage ranks them and the retrieval and in-context results follow the ranking; within one corpus, coverage alone does not reproduce it.**

7.7 Where the embedding is a proxy

Every number above is computed in an embedding, and for two of our domains the embedding is a proxy for an artifact that lives somewhere else — a rendered image, a rendered instrumental. How much of the diversity the loop measures survives the rendering step, which embedder to steer through, and how to close the loop in the artifact's own space are the subject of the companion paper, *Proxy Embeddings: Steering and Measuring Diversity Through a Space That Is Not the Artifact's*. Three of its results bear directly on the claims here. Text-embedding similarity between image instructions predicts rendered-image similarity at Pearson r between 0.167 and 0.421 across the seven rendered arms — 14.5% shared variance pooled, 8.2% within-arm — so a text-side diversity method, ours included, is optimizing a proxy that leaves most of what the reader receives unexplained; the `rac+vision` arm of §7.2, which renders a bounded sample and steers in CLIP space, is the method applied one level down, and it is the best arm in that table (companion §7.6). In audio, whether a prompt can be steered before rendering is a property of the *embedder* — prompt-to-track alignment 0.68 under MuQ-MuLan against 0.18 under CLAP-music — and per-arm diversity verdicts flip between embedders, so no audio-diversity number should be published without naming its embedder (companion §7.2). And conditioning loses roughly a third of its grip at the seam where the model that proposed the axes hands its instruction to a model that never saw them (§8.1; companion §7.7).

8. What makes it work: axis-level evidence

The results above evaluate corpora. None of them asks whether a *single axis* did anything, and the axis scoring assumes throughout that a commanded level shows up in what comes back. This section audits that assumption, and the audit turns into the most transferable part of the work: two things a realization check has to measure, one budget the conditioning interface has that nobody models, and one rule about what an axis must enumerate to reduce a corpus's repetition rather than relocate it.

8.1 Do the axes reach the artifact?

The short answer, established by intervention rather than by correlation, is that **the axes are levers and not labels**: holding a base spec fixed and setting one axis to each of its levels moves the poem substantially on prosody, and moves the written image instruction on nine of eleven axes. The longer answer is that how much of that survives to the finished artifact depends on one structural fact — whether the model that makes the artifact is the model that proposed the axes — and that the observational audit everyone would reach for first cannot establish either, because it is confounded in a way corpus-sized data cannot fix.

The test is a manipulation check run on the artifact, never on the prompt. For each axis we ask two questions with independent instruments. Is the commanded level *identifiable* — embedding each level description with CLIP, for an image generated under level ℓ , is $\text{CLIP}(\ell)$ the nearest of that axis's levels, against a chance rate of $1/|\text{levels}|$? And does the axis *separate* the artifacts at all — η^2 of the artifact embeddings grouped by commanded level, with a permutation null over shuffled labels. The second is weaker and assumption-free: it does not require the picture to match the words, only that the levels produce systematically different pictures.

Both instruments live inside the embedding the method optimizes, which makes agreement partly circular, so we add a third channel that does not: pixel statistics for images, and **prosody** for text. The text case matters more, because nomic-embed-text is nearly blind to metre, rhyme and line length, so an axis can look realized in embedding space while changing nothing about the poem as *verse*.

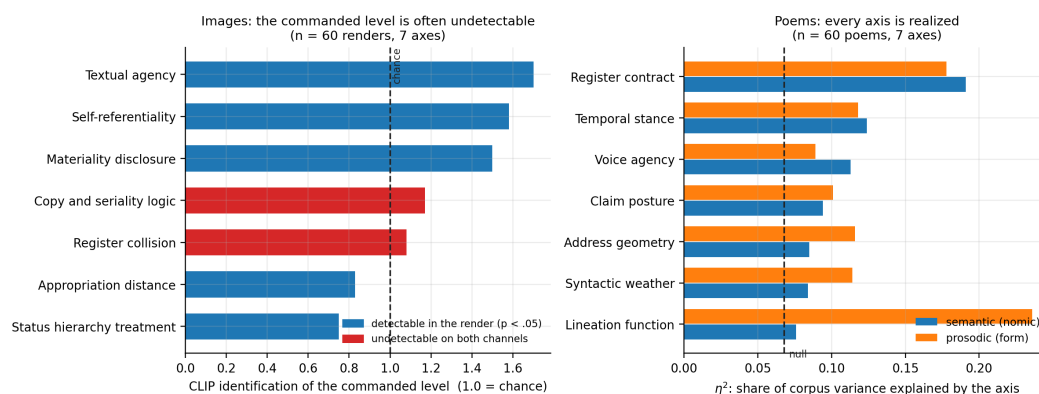


Figure 8. Axis realization measured on the artifact. Left: on 60 max-min renders, three of seven axes identify their commanded level at or below chance, and two are undetectable on both the CLIP and pixel channels. Right: on 60 poems, every axis is realized — on the semantic channel, the prosodic channel, or both. Dashed lines mark chance and the permutation null.

The observational audit splits the two domains sharply. On the 60-render max-min corpus, mean identification is $1.23\times$ chance; three of seven axes identify their level at or below chance, and only three of seven move pixel statistics at $p < 0.05$. The axis the *attribution* scoring ranked **first** — *Copy and seriality logic* — is the least realized axis in the corpus on both channels ($p = .674$ on CLIP, $p = .920$ on pixels), and under manipulation scoring it ranks last; rank agreement with measured realization moves from -0.429 to $+0.679$. This is §5.3's inversion in the field rather than in simulation. On the 60-poem pilot the same audit finds all seven axes realized at $p < 0.05$ on at least one channel, with effect sizes roughly twice the image corpora's.

The observational audit is also confounded, and cannot be de-confounded at these corpus sizes. It reads a corpus in which the axes co-vary, so an axis's apparent effect is entangled with every axis it travels with; adjusting for that needs a dummy per level per axis, and with nine to eleven axes at five levels each a 50-poem corpus has fewer rows than columns. The manipulation is available where the adjustment is not. Holding a base spec fixed, we set one axis to each of its levels in turn, generate, and read the displacement — centring within base so that only variation produced by *setting* the axis contributes, with a null that shuffles levels within each base.

Poems, by intervention. Across two independent runs of 440 generations, four axes are realized at $p < 0.05$ in both: *Register contract* ($p = 0.277$), *Temporal stance* (0.240), *[form] Poem scale* (0.170) and *[form] End-rhyme architecture* (0.112), on **prosody** — **a channel the objective never optimizes**. These are effects in isolation, not in a live run: the probe removes the avoid-block and the local negatives, which contribute variance no axis controls, so the same axis accounts for a smaller share of a running corpus than it does here. What the probe establishes is the direction the observational audit cannot. Setting an axis moves the poem.

Images, by intervention. The image pipeline has two stages — the model that proposed the axes writes an instruction, and gpt-image-1-mini renders it — so the same manipulation can be read on both sides of the handoff. Over all eleven axes at three bases and five levels, nine of eleven axes are realized in the **written instruction** at $p < 0.05$ (mean $p = 0.148$), against four of eleven in the **render** (mean $p = 0.096$); ten of the eleven attenuate, a mean loss of 35% of the text-side effect (Wilcoxon signed-rank $p = .005$). The conditioning is there, in the prompt, and measurable; what fails is its survival through a generator that never saw the axis set. The companion paper on proxy embeddings takes that seam as its subject (its §7.7): it separates the renderer's loss from the instrument's, since the four perceptual axes attenuate most in CLIP and are exactly the axes CLIP is least equipped to register, and it draws the consequence for the modality-agnostic claim — it holds for the calculus, which needs only an embedding, but not the conditioning, which needs a generator that reads the language the axes are written in.

8.2 Expansion: naming what a corpus never varies

Reading the artifacts says the same thing the statistics do, and says it faster. Eight max-min renders spaced evenly through a sixty-item run contain four instances of one conceptual gag — ironic text-in-image — and are otherwise uniformly frontal, centred, flat-depth, evenly lit and saturated. Absent from all eight: photography as a medium, monochrome or restricted tonal range, shallow depth of field, directional or low-key lighting, atmospheric or linear perspective, off-centre composition, and any viewpoint other than eye-level. The poems fail the same way along a different family: every one of sixty is roughly fifteen lines of mid-length free verse spoken by a lyric *I* in one elegiac register, with coefficients of variation of 0.08 on lexical richness, 0.16 on line count and 0.26 on syllables per line. *Lineation function* is the only formal axis, has the **largest** prosodic effect of any axis ($\eta^2 = 0.236$), and still moves line length by only a quarter of its own mean — because all five of its levels are free-verse variants.

Both are the §5.4 limit seen from two domains, and that the missing family is *optics* for images and *prosody* for poems is not a coincidence: in both cases it is the dimension the artifact has and the axis-proposing prompt does not think to ask about, having been asked for axes of variation in the abstract rather than in the thing being made.

The expansion step is aimed exactly here, and its proposals are checkable against that reading. Shown twelve renders and told which axes already exist, the judge returns *viewpoint and perspective*, *directional lighting*, *shot scale* and *motion and blur*, with concrete levels for each — the dimensions the reading named as missing, recovered independently from the pictures alone. Shown ten poems, it returns *poem scale* (two lines, 3–8, 9–24, 25–100, 100+ in titled sections) and *prosodic grid* (free verse, syllabic, accentual, accentual-syllabic, fixed metre with end rhyme).

Poems: every form dimension opens. At matched n , coefficient of variation on the features the baseline held nearly constant, each expanded arm scored against its own token-matched control, three seeds, 50 poems per arm, both conditions from one generator at one commit:

form feature	seed 7	seed 11	seed 23	mean
lines per poem	11.0×	12.6×	3.5×	9.0×
mean line length	1.7×	4.8×	1.8×	2.8×
syllables per line	1.6×	5.3×	1.7×	2.8×
syllable-count regularity	1.5×	2.2×	1.9×	1.8×
rhyme density	1.6×	1.2×	1.1×	1.3×
type-token ratio	1.9×	0.9×	1.1×	1.3×

Poem length opens in all three seeds — the ranges run (2, 115), (1, 300) and (2, 32) against controls of (12, 20), (12, 27) and (14, 22) — and the two features tied most directly to line construction rise in all three. The effect is concentrated there: rhyme density moves $1.1\times$ to $1.6\times$ and the type–token ratio is flat, varying between $0.9\times$ and $1.9\times$ with no consistent direction. **The generation budget is not what opens the range.** Long poems need tokens, so every arm is scored against a control at the same 2,500-token cap with expansion disabled; that control writes poems of 12 to 20 lines at a line-count coefficient of variation of 0.112, against 0.117 for the same generator capped at 700 — four times the budget, no instruction to use it, and the same length of poem. What opens the range is being asked for two lines, or for a hundred. And **semantic diversity rises alongside form diversity rather than paying for it:** centered Vendi $38.81 \rightarrow 40.84$ and median NN distance $0.2393 \rightarrow 0.2976$, a 24% gain, in the same embedding at the same matched n .

Reading the poems is again the fastest check. The expanded corpus contains a two-line epigram — *"Had I filed the silence—no: the silence filed me. / Under review; we remain—perhaps, I remain."* — a sixty-line numbered sequence, a bureaucratic intake report, a poem that arranges itself spatially on the page, and rhymed metrical quatrains: *"We thought the road would hold us fast, / We kept our names inside the rain, / We said the worst had surely passed, / Or so we told ourselves again."* That is iambic tetrameter in ABAB, and **it appears nowhere in the baseline sixty at any budget**, because nothing in the elicited axis set named metre or rhyme. Which absent family the judge names first is itself seed-dependent — scale and a combined prosodic grid at one seed, metre and rhyme as separate axes at the other two — so the effect is better described as *expansion opens the formal dimensions the axis set omitted* than as a claim about any particular feature.

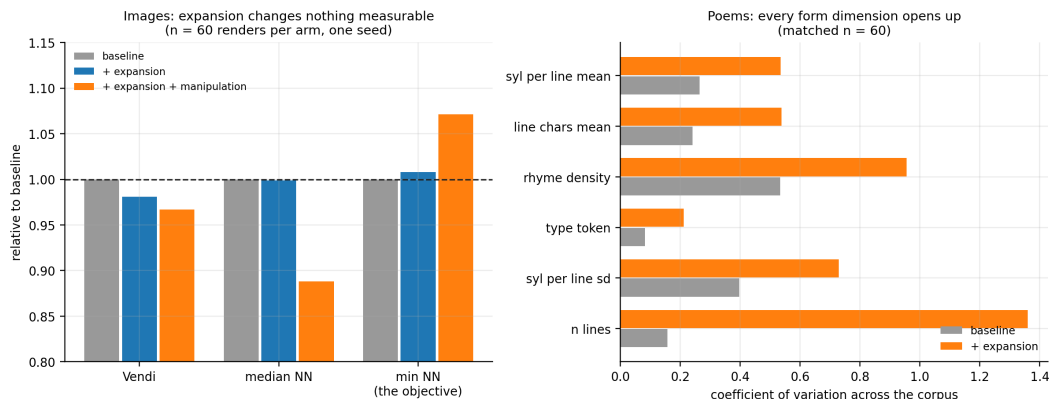


Figure 9. The same expansion mechanism in two domains, at matched budget. Left: three image arms scored relative to the baseline; only min NN distance, the quantity the arm maximizes, improves, and no difference exceeds what one seed can establish. Right: coefficient of variation on poem form features, baseline against expanded — every dimension the baseline held nearly constant opens up, line count by a factor of nine.

Images: no measurable effect, limited by a budget nobody models. Three arms of sixty renders, one seed, identical seeded axes, differing only in the mechanism under test, give centered CLIP Vendi 43.81 (baseline) against 42.97 (+expansion) and 42.37 (+expansion, manipulation-scored); only min NN distance — the quantity a max-min arm maximizes — improves, by 7%. This is despite the added perceptual axes being the *best-realized* axes in the run (mean $\rho = 0.036$ against 0.016 for the seeded axes) and the renders visibly acquiring optical modes the baseline never produces: an extreme macro, a radial-zoom blur, a wide vista with atmospheric depth. At one seed and $n = 60$ a 3% Vendi gap is inside run-to-run variation, so this is a null rather than a ranking.

The explanation is that conditioning on more axes means more contracts in one prompt, and per-axis obedience falls as that load rises. Four image arms spanning 7.0 to 11.0 contracts per item, same budget and same seeded axes:

arm	axes	contracts	per-axis ρ	palette	min NN	Vendi
baseline	7	7.0	—	0.417	0.1540	43.81
+ expansion, manipulation-scored	10	8.7	0.0219	0.533	0.1650	42.37
+ expansion	10	9.5	—	0.500	0.1552	42.97
perceptual axes pre-seeded from item 0	11	11.0	0.0126	0.667	0.1313	42.39
subset conditioning, 7 of 11	11	7.0	0.0167	0.500	0.1728	43.04

The *palette* column is the fraction of renders sharing a palette with a nearest neighbour, and it rises with contract load, from 42% at seven contracts to 67% at eleven, and mean realization halves between the two arms where it is measured. The fourth row is the strongest form of the expansion test — the four perceptual axes in force for every item rather than arriving mid-run, so each conditions 100% of the corpus instead of 7% to 80% — and it is the worst arm on every structural measure. **An axis added late is diluted by the items generated before it; an axis added early dilutes every other axis in the prompt.** There is a contract budget here, and past roughly seven simultaneous contracts the marginal image axis costs more in compliance than it returns in reach.

The budget belongs to this conditioning interface rather than to conditioning in general, and the boundary is informative. The interventional probe of §8.1, run at seven contracts and at eleven, finds per-axis obedience **flat** in the load: across three runs the mean difference between loads is +0.002, −0.036 and +0.044, changing sign and swamped by its own spread. An image prompt carries contracts, mined attractors, structural bans and an overused-word list into a generator that never saw the axis set; a poem prompt carries contracts into the model that proposed them. Where the conditioning does not cross that seam, seven contracts and eleven are equally well honoured.

The budget binds the prompt, not the lattice, and nothing requires every axis to appear in every spec. Drawing seven contracts per item from the full eleven-axis set, least-used-axis-first so each axis accumulates equal usage across the run, keeps the corpus's reach while holding prompt load where compliance is highest. Against the pre-seeded arm — the same eleven axes, four fewer contracts — min nearest-neighbour distance rises 31.6%, per-axis realization 33%, and palette sharing falls a quarter; against the baseline, at the same prompt load but drawing from a lattice 1,300 times larger, min NN is 12.2% higher, the best value of any arm here. Centered Vendi sits slightly below baseline and palette sharing slightly above it, so this is a gain on the objective rather than a sweep, and at one seed per arm on the noisiest statistic in this work the magnitude should be read as approximate. What is not approximate is the design point it makes concrete: **the lattice describes what the corpus can reach and the spec describes what one prompt must carry, and holding the second fixed while enlarging the first is what lets expansion pay.**

8.3 Completeness: enumerate the whole space, or the concentration moves

Expansion answers *what is missing*, which is the right question when the missing family is unknown. It does not answer *what must be covered*. Where a target partition is specified in advance, the distinction decides whether an intervention helps or backfires, and following it to its end produces the most transferable rule in this paper.

An exam item bank is built to a **blueprint** — content area crossed with cognitive level — so unlike optics or prosody it has a target partition specified in advance, and coverage of that partition is what a psychometrician grades. The elicited axes span none of it: all seven describe item architecture, and the elicitation could not have proposed otherwise, since it asks for axes of item-design variation and excludes topic lists. The consequence is visible on the page — three items drawn at random are all cloud backup and availability-zone problems — and in the corpus, where one blueprint area takes 35% of items. Naming breadth in the generation prompt does not help; that prompt already names architecture, storage, networking, security, cost and reliability.

Expansion behaves exactly as it does elsewhere and does not fix it. Shown twelve items, the judge recovers three content dimensions the bank barely varies — deployment and locality model, service

abstraction level, data modality — and they are real dimensions, recovered from the artifacts alone. Conditioning on them makes the bank **lumpier**: under a judge that classifies each item into one blueprint area, blind to arm, the dominant area's share *rises* from 35% to 47% and normalized entropy falls from 0.700 to 0.645. The three axes are content-*adjacent* rather than blueprint-*aligned*: "streaming, analytical, time-series data" is a genuine axis of variation, and it drives items into one area rather than across ten.

A single axis whose levels *are* blueprint areas does spread it, and how much depends on whether it names all of them.

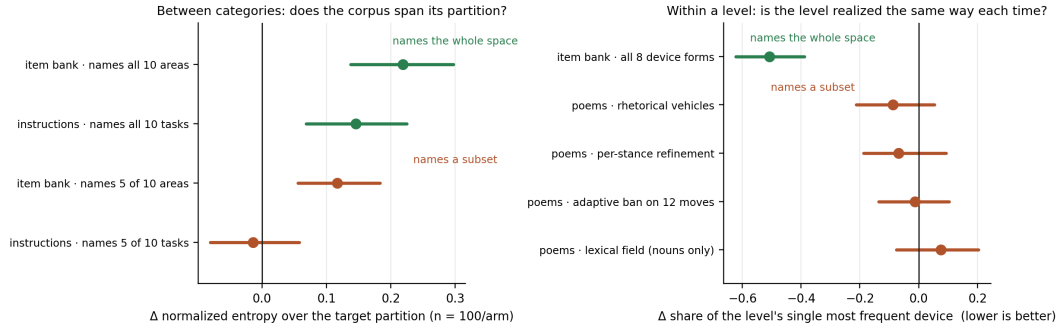


Figure 10. The completeness rule, nine controlled comparisons with blind judges. Left: axes that enumerate a target partition, scored on normalized entropy over that partition ($n = 100$ per arm, bootstrap 95% CI). Naming half a partition helps a corpus that is failing to span it and does nothing to one that already spans; naming all of it helps both. Right: axes that enumerate the ways a single level can be realized. The item-bank row is scored by a blind judge on the share of items using the dominant device form; the four poem rows are scored lexically, on the share of each Claim posture level's poems using that level's single most frequent non-echo word ($n = 80\text{--}90$ per arm). Enumerating every device form of one construct removes half the concentration; every intervention that names only part of the realization space moves it somewhere else instead.

An axis naming the **five underused** areas takes the item bank's normalized entropy from 0.700 to 0.816 (+0.117, 95% CI [+0.056, +0.183]), the dominant area's share from 35% to 19%, and the Gini coefficient over areas from 0.606 to 0.410; the three arms' distributions differ at $\chi^2(18) = 162.6$, $p = 3 \times 10^{-25}$. An axis naming **all ten** takes it from seven areas to ten and from 0.700 to 0.926 (+0.219, [+0.138, +0.297]) — nearly double.

Instructions show why the slice is the weaker instrument. Their partition, the taxonomy of what a user asks an assistant to do, is already spanned by axes describing a request's *shape*: six such axes alone reach nine of ten task categories at entropy 0.811, flatter than the item bank managed with any axis. Adding an axis naming five underused categories moves an enormous amount of probability mass — those five rise from 14% of the corpus to 81% — and changes total spread not at all (−0.014, [−0.080, +0.058]), because the five it does not name are driven out: *decide* from 27 instructions to none, *plan* from 23 to one. An axis enumerating **every** category has no unnamed half to trade away, and level balancing then commands each equally often: entropy 0.811 → 0.959 (+0.146, [+0.069, +0.225]), all ten categories reached, the largest taking 25%. So the prescription is unconditional and needs no prior audit of where a corpus is thin — **where a partition must be covered, an axis should name all of it** — and a slice is worth reaching for only when the goal is to *move* a corpus toward particular categories rather than to spread it.

The same rule holds one level down, and there it exposes a defect no realization audit can see. Every audit in §8.1 asks whether a commanded level *moves* the artifact, and every one computes that *between* levels. A collapse *within* a level is invisible to all of them. The psychometric bank shows one. Its Irrelevant-information treatment axis is plainly obeyed — items told to carry no irrelevant detail use a colour word 7% of the time, against 52% to 62% for items told to carry some — and yet half of the 1,999-item corpus contains a colour word, because the red herring is nearly always a coloured card, a star, a printed dot. Generating afresh reproduces it: under a judge

that names the *form* the irrelevant detail takes, blind to arm, 76% of items use a physical-appearance detail and two of eight available forms never appear, for a normalized device entropy of 0.425 — far below the between-category spread of any corpus measured here. This is not cosmetic. Irrelevant information functions only while the examinee cannot recognize it by its form; a bank whose red herring is always a colour sentence teaches candidates to skip colour sentences, and the distractor stops measuring anything.

The remedy is the same prescription applied inside a level: an axis enumerating the forms a red herring can take, rather than trusting a level description to vary its own realization. It takes device entropy from 0.425 to 0.933 (+0.504, [+0.385, +0.626]), the dominant form from 76% to 24% (−0.507, [−0.622, −0.389]), and all eight forms into use, **while both arms keep the contract the axis specifies**: in each, 100% of items still carry detail whose removal would not change the answer. **Realization therefore has to be audited twice — once to establish that a level moves the artifact, and once to establish that it does not move it the same way every time.**

The defect is general and the poem corpora carry it more heavily. Scoring each level by the share of its items using its single most frequent non-echo content word, poems average 0.533 against 0.412 for the psychometric bank, with a worst level at 0.811. *Claim posture: ritual pronouncement* is realized through one liturgical formula — "By ash, by antenna, by the names", "By the gathered breath", "By salt and smoke, I seal this house" — drawing on a fixed stock of ash, salt, bells and thresholds across independently generated poems; *Syntactic weather: nested subordinate clauses* reaches for *because* in 79% of its items, and *Register contract: ceremonial elevation for silence* in 72%. The measure cannot distinguish a stereotyped device from a word the level entails, and *dialogue with conflicting claims* scoring 0.73 on *says* is the latter, which is why the levels above were read rather than only counted.

Three attempts to transfer the remedy fail in the same shape, and that is what identifies the rule. A red herring's forms are a closed list; a poetic stance can be delivered by any rhetorical means. Enumerating *vehicles* instead — an imperative, an unanswered question, a negation, a bare object, a reported speech act, a conditional, an inventory, a self-correction — is orthogonal to which stance is carried and composes with every stance axis. It lowers device share on **all seven** poem axes at once, 0.618 → 0.570 (7 of 7, one-sided sign test $p = .008$; the aggregate difference is −0.048, [−0.101, +0.012]) while the commanded posture is still exhibited in 96% of poems against 95% without it. But the *size* is the point of contrast: an axis enumerating the realization space of a specific construct moved an item bank's dominant form from 76% to 24%, where a general vehicle axis moves poems by 0.04 and leaves *ritual pronouncement* reaching for *let* in 69% of its poems, down from 88%.

Making the poem axis construct-specific does not close the gap. Refining each posture into five ways *that* stance is performed — ritual pronouncement into invocation, blessing, oath, anathema, consecration — gives the largest single-level improvement of any arm (*let* falls to 56%) and the highest contract adherence (99%), and does not lower device share overall (0.595 against a 0.662 baseline, −0.068 [−0.188, +0.093]) because *name* becomes the dominant word in three of five levels. Nor does making the ban adaptive: rebuilding it every batch from a judge's reading of the rhetorical *moves* the corpus has just started overusing — the register the defect lives in — leaves device share at 0.650 (−0.012, [−0.136, +0.104]), with *let* falling from 88% to 69% for ritual pronouncement and *through* arriving at 69%. Twelve banned moves is still a subset.

The clearest case is the one that makes the concentration **worse**. Commanding the poem's concrete nouns from one of eight lexical fields is obeyed almost perfectly — the commanded field's vocabulary appears in 98% of poems and is the dominant field in 94%, against a chance rate of 12.5% — and device share nevertheless *rises*, 0.662 → 0.738 (+0.076, [−0.076, +0.204]), with the posture contract holding at 96% and craft unharmed. The words carrying the new concentration are the ones the axis left free: *confess* in 81% of confessional poems, *mine* in 88% of refusals, *beneath* in 75% of flat assertions. Constraining the nouns concentrated the verbs and pronouns.

The rule that explains all seven comparisons without appealing to domain is therefore about *completeness*:

An intervention that enumerates a proper subset of the space a measure is defined over displaces concentration into the complement; only enumerating the whole space reduces it.

intervention	enumerates	effect on the measured concentration
5 of 10 task categories	subset	unnamed half 86% → 19%; total spread flat
all 10 task categories	whole	spread up in a lumpy corpus and a flat one
all 8 device forms of one construct	whole	dominant form 76% → 24%
rhetorical vehicles (partial lexical control)	subset	−0.04; mass moves to a new word
per-stance device refinement	subset	flat; <i>name</i> becomes dominant in 3 of 5 levels
adaptive ban on 12 rhetorical moves	subset	flat; <i>let</i> → <i>through</i>
lexical field over nouns	subset	obeyed 94%; mass moves to verbs and pronouns
ban on the 8 most overused words (images)	subset	+0.089; <i>light</i> → <i>illumination</i>
ban on the 8 most overused words (poems)	subset	+0.112; <i>let</i> → <i>beneath</i> , <i>answer</i> → <i>perhaps</i>
mined attractors, stated as prose	not an enumeration	−0.136, the largest reduction measured

One measurement keeps this from being read as conservation. On whole-vocabulary mean pairwise word overlap between poems sharing a posture, the lexical arm has the **lowest** overall repetition of any arm (0.0387, [0.0363, 0.0412] against the baseline's 0.0452) at the same time as one word becomes more dominant within it. Both are true — the field axis spreads the nouns across the corpus while the posture's stock verb tightens — and total overlap varies across arms by 14%, so repetition is not a fixed budget that conditioning merely shuffles. What conditioning reliably does is leave concentration wherever the axis set does not reach. That is the argument for auditing realization in a channel the objective never optimizes, for measuring it more than one way, and for reading the artifacts rather than only the scores.

The rule reaches the pipeline's own machinery, where the two blocks that stand against concentration pull in opposite directions. Image instructions written from the contracts alone are dominated by *mixed-media* on nine of eleven axes, a within-level concentration of 0.564. Adding the hard block — the structural bans together with a list of the corpus's eight most overused words — raises that to 0.653, and adding both blocks raises it further; adding the mined-attractor block alone lowers it to 0.427 and leaves no dominant device behind. All four arms wrote 110 of 110 instructions, and every device is scored with the vocabulary of each arm's own prompt blocks excluded, so no arm is credited for repeating its instructions back.

What replaces what shows the mechanism at word granularity. The hard block bans *light*; *illumination* becomes the dominant device on fifty of the level-slots measured. Banning a word leaves the concept, and the generator reaches for its nearest synonym. A prohibition names a subset by construction — that is what a prohibition *is* — so it relocates concentration rather than reducing it, exactly as an axis does. The attractor block is the one mechanism measured here that does not: it describes a *tendency* in prose instead of enumerating tokens, and so offers no complement for the mass to move into.

The effect replicates in poems, where the substitutions are legible one by one. Banning the eight words the corpus most overuses — *still*, *hand*, *name*, *record*, *rain*, *let*, *say*, *silence*, seven of them already the dominant device of some level — raises device share from 0.525 to 0.637 (+0.112, 95% CI [+0.016, +0.180]) when both arms are scored on the vocabulary available to both, and every level rises: *record* gives way to *because*, *answer* to *perhaps*, *mine* to *cannot*, *let* to *beneath*. The commanded posture survives in 98% of poems, so this is displacement rather than a broken contract, but craft falls to 5.97 from the 6.5 to 6.7 of the other arms. Prohibition at the token level costs quality and buys concentration, in both domains where it has been measured.

Taken together the interventions grade by what they name rather than by domain: banning tokens makes concentration worse, banning tendencies leaves it flat, and enumerating a whole space is the only thing that reduces it.

8.4 What best-of-K buys depends on what the selector optimizes

§4.2 shows that best-of- K oversampling improves the expected min-gap only as $K^{1/m}$, and that result assumes selection *on the gap*. Our own loop does not select that way: it keeps the candidate maximizing $0.40 \cdot \text{craft} + 0.35 \cdot \text{orthogonality} + 0.25 \cdot \text{min-distance}$, a blend in which the quantity the corpus is scored on carries the smallest weight. The distinction is measurable, and it matters more than the weights suggest. Poem arms of 50 items at $K = 6$, three seeds, changing only the selection rule:

seed	min NN, blend	min NN, gap-only	5th-ptile NN, blend	gap-only	craft, blend	gap-only
7	0.1910	0.2510	0.2111	0.2538	8.54	8.17
11	0.1920	0.2168	0.1984	0.2413	8.55	8.16
23	0.2377	0.2788	0.2427	0.2796	8.47	8.26
mean	0.2069	0.2489	0.2174	0.2582	8.52	8.20

Selecting on the gap alone raises the minimum nearest-neighbour distance in **3 of 3 seeds**, by **+20.3%** on the means (range +12.9% to +31.5%; paired $t(2) = 4.13$, $p = 0.054$), and the fifth percentile by 18.8%. It costs 0.32 of a craft point on a ten-point scale, and the cost replicates more tightly than the benefit — 8.52 ± 0.04 against 8.20 ± 0.04 — which is what a utility term being removed should look like. The effect also explains a sweep that is otherwise puzzling: under the blended selector min NN is **not monotone in K** (0.1828 at $K = 1$, 0.2148 at 3, 0.1910 at 6), and under the gap-only selector it is (0.2278 at 3, 0.2510 at 6). Additional candidates do buy minimum gap, but only for a selector that spends them on it; the blend spends them on craft and bulk spread, both of which rise monotonically in K under either rule.

The practical statement is narrow and easy to get wrong: **a corpus scored on min-gap and selected on a quality-weighted blend is being oversampled for something other than what it is scored on**. The $K^{1/m}$ ceiling is real, and a system can fail to reach even that ceiling by pointing its selector elsewhere. Three seeds at $n = 50$ is a small replication and the effect size varies by a factor of two and a half across them; the direction is consistent and the magnitude approximate.

The same question at the *render* has a sharper answer, because the two channels have very different effective dimensions. Best-of- K on the written candidate happens before the handoff and §8.1 locates the loss after it, so we also ran $K = 3$ renders behind each of 60 fixed instructions and kept the render maximizing min distance to the accepted set — once in CLIP, once in optical statistics, over the same candidates. Selecting on optics raises pixel min NN from 1.039 to 1.292 (1.24 $\times$) and the fifth-percentile NN by 0.336 (paired subsample, 95% CI [+0.128, +0.488]), **and costs nothing in the channel the method optimizes**: CLIP min NN moves by -0.001 with the interval $[-0.029, +0.033]$ tight around zero. Selecting on CLIP raises CLIP min NN by 0.040 ([+0.025, +0.071]) and moves optics not at all detectably. Both gains lie on the $K^{1/m}$ curve — 1.24 $\times$ at $K = 3$ implies $m \approx 4.4$, and 1.04 $\times$ implies $m \approx 17$ — so render-side selection is oversampling rather than a new mechanism. What the dimensions add is an allocation rule: oversampling buys far more in a low-dimensional channel than in a high-dimensional one, and **if a pipeline is going to pay for extra renders, it should select them on the channel the objective cannot see**.

9. Limitations

- **One embedding oracle.** All geometry is nomic-embed-text geometry with cosine distance, on both objectives. Whether *diverse* or *covered* under one embedder transfers to another is the subject of the companion paper on proxy embeddings, which measures text-to-CLIP correlations on rendered outputs of only $r = 0.167$ to 0.421 and shows per-arm audio diversity verdicts flipping between embedders, and which shows the *ranking* of methods surviving four independent representations. For any domain with a downstream rendering step the objective should be defined in the space of the rendered artifact. Here it was not.
- **The head-to-head is not like-for-like.** Our winning arm reads a sample of the target distribution as embeddings on the steering side; the released corpora had no such input. §7.3 states

the like-for-like rows explicitly and the claim is scoped to targeted generation. Separately, matched- n sampling controls for size and cannot control for the model that wrote the items: the released corpora are 20–70 $\times$ larger, written by different generators, with human curation in at least one case.

- **ϵ is chosen, not learned.** Every covering guarantee is stated per- ϵ , and our calibration from quantiles of reachable nearest-neighbour distance is a heuristic; §4.6 shows what optimizing at the wrong radius costs. A multi-resolution objective, integrating coverage over a prior on ϵ , is the obvious next step.
- **The pool is the measure.** Coverage is relative to the reachable pool throughout. Behaviours the base generator cannot emit are invisible until refinement opens them, and the pool refresh is only as good as the refinement trigger that fires it.
- **Judge bias is unmeasured.** Language-model judges prefer fluent, typical text, which is the bias that would work against a diversity system. We use a judge for craft, validity, blueprint area, device form and posture, and did not calibrate any of them against human raters. The exam-item enemy radius is likewise adjudicated by a model under a psychometric rubric rather than by a credentialed psychometrician, and δ is specific to this embedder and this item type, with a bootstrap interval that includes zero.
- **Additive spec composition.** The axis scoring treats conditioning attributes as composing additively in embedding space, which is what makes a spec's position predictable before it is generated. Real interactions between prompt attributes are not additive, and the partial-effect estimator of §5.3 leans on that assumption harder than the original scoring did. The interventional probe is the version that does not assume it, and it costs generations.
- **Observational realization is underpowered and partly unidentifiable.** The audit groups sixty items into five levels per axis, which after the null correction leaves little room to separate a small real effect from none; and at nine to eleven axes a partial-effect design needs more columns than a 50-item corpus has rows. We can say the image axes are much less realized than the poem axes, and that particular axes are inert; we cannot put a confident number on how much of the image corpora's diversity the axes contribute.
- **Several mechanism results are single-seed.** The image arms are one seed each at $n = 60$, which is why §8.2 reports the image expansion as a null rather than a ranking and why the subset-conditioning margin is reported as approximate. The poem expansion and the selector result are replicated across three seeds; the completeness comparisons are single runs at $n = 80$ –100 per arm with bootstrap intervals.
- **Expansion is proposed, not verified, by the same kind of model.** The step that asks what a corpus never varies is a language model looking at artifacts. Its proposals matched an independent reading of the same corpora in three domains, but a blind spot shared between that judge and the axis-proposer would be invisible to us in exactly the way the original missing families were.
- **An axis that changes the artifact's scale breaks fixed budgets downstream.** Expansion is the one operation here that changes the *scale* of the artifact rather than its character, which couples it to every fixed token cap between the spec and the scored output. The method does not currently represent that coupling.
- **The temperature arm is unverified.** `high_temp` requested a temperature the generator's end-points do not list as supported (§6), so its rows may replicate the default configuration. Every statement about temperature in §5.6 and §7.1–7.4 rests on those rows and carries the same doubt.
- **Quality is scalar.** Craft is not one number, and collapsing it to one lets a system trade away dimensions of quality the judge does not score.

10. Conclusion

No measure here is limited by its optimizer. What determines whether a corpus can keep growing without collapsing into paraphrase, and whether it can blanket the space it is drawn from, is the dimension of the generator's conditional support relative to the manifold it lives in. Saturation arrives at rate $n^{-1/m}$ rather than $n^{-1/d}$, a single prompt covers an ϵ^{d-m} fraction of what the model could

write, and only transverse prompt motion raises the ceiling. Everything else buys a constant factor against an asymptotic problem.

Given an embedding oracle and no inverse, the system cannot compute its way to the next item; it can only choose what to condition on. Recursive Axis Conditioning is our answer, and it works: first of twelve corpora against released instruction sets with a twentieth as many items, an exam bank with no enemy pair at a judged radius where naive prompting yields 15% of nominal capacity, and separation from every baseline in three artifact domains measured on channels the loop never optimizes. Its most valuable output is the refine signal — the moment the scoring reports that nothing available is transverse any more is the moment the horizon has been reached, and the only remaining move is to ask the generator to subdivide its own vocabulary of variation.

The lessons that follow transfer past the method.

Name the measure, because the two classes are not one problem. Greedy k -center wins min-gap in both domains and finishes last on coverage; orthogonalized conditioning, load-bearing under max-min, scores below plain conditioning under coverage. Pointing the loop changes exactly two things, how an axis is scored and how a candidate is chosen, and those two are enough to invert which method looks best. A corpus is diverse *with respect to* an objective.

Score your control variables by manipulation, not by attribution. Level vectors read off a corpus in which every axis varies at once are marginal means, and because an effective axis fills the corpus with variance along its own direction while transversality credits variation lying outside the occupied span, the resulting score penalizes an axis in proportion to how well it works. On data where the answer is fixed by construction the observational form ranks the inert axis first in most runs; partial effects and a null-corrected realization factor recover the true order. The lesson generalizes to any system that scores its own control variables against its own outputs.

Name the whole space, or the repetition moves. Conditioning does not so much remove a corpus's tendency to repeat itself as relocate it into whatever the axis set does not name. An axis over half a task partition drove the unnamed half from 86% of a corpus to 19% and left total spread unchanged; an axis over all of it raised spread in a lumpy corpus and a flat one. An axis over every form a red herring can take moved an item bank's dominant device from 76% to 24%; axes over part of a poem's realization space moved it by 0.04, or made it worse while being obeyed 94% of the time. The corollary is a practice, not a theorem: realization has to be audited twice, once between levels and once within one, because a level that is obeyed every time and performed identically every time passes every check in the literature.

Two things bound all of it, and both are measured in the companion paper on proxy embeddings. The first is a seam: conditioning reaches the written instruction at mean $\rho = 0.148$ and the render at 0.096, so a third of its grip is lost exactly where one model's language becomes another model's input, and where that seam exists axis realization has to be measured rather than assumed. The second is sharper. Ninety percent of our generated exam bank's answer keys sit in option position A — a bank an examinee can score 90% on without reading a stem — and not one measure in this paper detects it, because key position carries no semantic content for an embedding to encode. A diversity objective defined over an embedding cannot see any property the embedding does not encode, including the ones that decide whether the artifact is usable at all.

Which is the argument for the practice we ended up recommending, and it is short. Count your duplicates before computing anything else. Report the literal and the latent separately, and name the denominator. Check that the knobs you are turning are connected to anything, by intervention rather than by correlation. Measure at the level of the artifact you are shipping, and audit in at least one channel your objective cannot see — it is where we found both our strongest result and our largest undetected defect. And read the output, because a corpus can be diverse along every dimension you thought to name and uniform along the one you did not.

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